GSE 518 • PROBABILITY AND STATISTICS FOR ECONOMETRICS

Cal Poly MSQE • Prof. Carlos A. Flores • Final Exam, Monday August 17

The Final Survival Guide

Everything from Set Theory through Asymptotics, rebuilt from the ground up in plain language, with the statistics half taught as seven repeatable machines.

SESSION 1 3 hours • target 80% • the five machines that produce most of the points
SESSION 2 5 hours • target 100% • depth, traps, the real Sample Final, a full mock
CHEAT SHEETS 2 pages, front and back • ordered by exam probability

Built from Notes I through XIV, Problem Sets 1 through 10 with solution keys, the Sample Questions from Old Final Exams and its answer key, the two Review handouts, and Dr. Flores's exam-information email of August 2026. Every numerical answer in the practice sections has been symbolically verified.

Read this page first

You told me the lectures blurred together, that you attended everything but froze when you looked at a problem because you could not remember what the symbols meant. That is a notation problem wearing a mathematics problem's clothing. It is extremely fixable, and this guide is built around that diagnosis.

The central claim: **this exam is not testing whether you can invent statistics. It is testing whether you can recognize which of about seven situations you are in, then run the matching machine without dropping a step.** Dr. Flores said as much on the midterm sheet: using formulas and processes correctly matters more than algebraic correctness. Partial credit lives in the setup.

EXAM INTEL FROM DR. FLORES

From his exam-information email, quoted in substance:

- The final is **about the same length as the midterm**: roughly 9 to 10 questions worth 10 to 11 points, with parts (a) through (e).
- **Hypothesis testing and confidence intervals are NOT on the exam.** Problem Set 11 is excluded entirely. The exam ends at asymptotics.
- You may bring **two pages**: the formula sheet you used for the midterm, plus a second one covering the second half of the class.
- Cheat sheets may contain **formulas and "steps" on how to solve problems**. **No examples and no exercises.** He is explicit that if examples were allowed he would have to ask harder questions.
- **You will be given any pdf, $E(X)$, or $\mathrm{Var}(X)$ you need.** You do not need to write the mean and variance of every distribution on your sheet.
- Consistent with the midterm, he does not plan to include "Hard" questions.

How the two sessions differ

SESSION 1 3 hours, target 80%	SESSION 2 5 hours, target 100%
The five machines that generate most of the exam: reading the notation, sampling distributions, the Method of Moments, Maximum Likelihood, and the asymptotic-distribution pipeline. Aim: never be confused about *what to do*.	Everything else, plus depth. The chi-squared/t/F construction machine, score and information identities, the CRLB three ways, UMVUE logic, MSE comparisons, Slutsky, the delta method, and every trap. Aim: never be wrong about *how to do it*.
Ends with a 12-item checkpoint.	Ends with the real Sample Final fully solved, then a fresh 10-question mock final with solutions.

Three rules for using this document

1. **Do not read passively.** Every worked example is written so you can cover the solution, attempt it, then uncover. Reading a solved derivation teaches you almost nothing. Failing at it for ninety seconds first teaches you a great deal.
2. **The decoder boxes are the highest-value content here.** If you have only one hour, read every amber decoder box and then the cheat sheets.
3. **Transcribe the cheat sheets by hand.** Copy mine, but write them out yourself. The act of transcribing is a third study session hiding inside a chore, and it is what will let you find things quickly under time pressure.

The blueprint: what is actually going to be on this thing

Dr. Flores gave an unusually precise map. Below, each "for sure you can expect" item is matched to the machine that solves it and to the block that teaches it.

What he said to expect	Source problems	The machine	Block
Construct a given distribution (χ^2, t, or F) from normal random variables	SF #5, PS7 #5	Standardize, square, add, divide by degrees of freedom	7
Find the MOM estimator	SF #3b, PS8 #1, #3	Set sample moments equal to population moments and solve	3
Find the MLE estimator	SF #4a, PS8 #2, #3	Log, sum, differentiate, set to zero, solve	4
Find the CRLB, decide whether an estimator is UMVUE	SF #4b–4c, PS9 #2	Score, then Fisher information, then $1/(nI)$, then compare	9
Asymptotic distribution of the MLE	PS10 #4b, #6a, SF #4e	The four-step pipeline	5
Show an estimator is consistent	PS10 #1, #2, #3b, #6b	Limit of bias is zero and limit of variance is zero	6
Compare two estimators by MSE	SF #2, PS9 #1	$\text{MSE} = \text{Var} + \text{Bias}^2$, compute both, compare	10
Delta method for a function of an estimator	SF #4g, PS10 #3b, #6b	Invariance to get the estimator, delta method for its variance	5, 11
Mean and variance of $\overline{X}$ and S^2, proofs	SF #1, PS7 #1–#2	Linearity of E, independence kills covariances	2
Score identity, information matrix identity	PS8 #4	Differentiate the log density once or twice, take expectations at θ_0	8

MEMORIZE

Notice the shape of the table. Six of the ten rows are the same four objects in different costumes: the **log density**, the **score**, the **Fisher information**, and the **sample mean**. Block 5 shows how a single pipeline answers four of these rows at once.

What is fair game from the first half

The exam is cumulative in *tools* even though the new material is all statistics. Dr. Flores's own note: expectation formulas show up a lot, and the midterm cheat sheet will be used. Concretely, from Notes I through IX you will keep needing:

- E and Var algebra: $E(aX+b)$, $\text{Var}(aX+b) = a^2\text{Var}(X)$, $\text{Var}(X) = E(X^2) - [E(X)]^2$
- Computing $E(X)$ and $E(X^2)$ by integrating against a pdf
- $\text{Var}(aX+bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X,Y)$
- Cov properties and bilinearity
- Independence implies zero covariance
- Standardizing a normal: $(X-\mu)/\sigma \sim N(0,1)$
- Transformations of random variables, change of variables
- Moment generating functions and their uniqueness
- Support and how it constrains everything

That is what Cheat Sheet 2 is for. Set theory, Bayes' rule and the joint-distribution machinery are unlikely to carry their own question, so they get a compressed refresher rather than a full treatment.

Master notation decoder

This is the page you asked for. Every symbol in the statistics half, what it *is*, and the single mistake people make with it. If you read nothing else, read this.

SYMBOL DECODER

The three thetas. This one confusion causes more lost points than any algebra error.

θ_0	**the truth**	The real, fixed, unknown value out in the population. It is a *number*, not a random variable. It never changes and you never observe it.
θ	**a candidate**	Any element of the parameter space Θ. This is the variable you differentiate with respect to and maximize over. Also a number, but a movable one.
$\widehat{\theta}$	**your guess**	The estimator. A function of $X_1, \ldots, X_n$, therefore **random**, therefore it has a distribution, a mean, and a variance.

Why it matters: the score identity says $E[S(\theta_0, X)] = 0$. It is false at a general θ. When a problem says "verify the score identity," it is telling you to plug in the *true* value and use the *true* moments $E(X) =$ whatever the distribution says. Every identity in this course holds only when evaluated at θ_0.

SYMBOL DECODER

The bar, the hat, the tilde, and the two sample variances.

$\overline{X}$	$\frac{1}{n}\sum_{i=1}^{n} X_i$, the sample mean. Random. This is the workhorse of the entire second half.
μ or $E(X)$	The population mean. Fixed, unknown, non-random.
S^2	$\dfrac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$. The *unbiased* sample variance. Divides by $n-1$. This is the one from Notes X.
$\widetilde{S}^2$	$\dfrac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2$. Divides by n. This is the one that shows up as the **MOM and MLE estimator** of σ^2, and it is *biased*.
$\widehat{\ }$ hat	Usually "the MLE" or just "an estimator."
$\widetilde{\ }$ tilde	Usually "a second, different estimator" being compared against the first, or the MOM one.

The relationship: $\widetilde{S}^2 = \frac{n-1}{n}S^2$. So $E(\widetilde{S}^2) = \frac{n-1}{n}\sigma_0^2$, which is smaller than σ_0^2: biased downward, but the bias vanishes as n grows.

SYMBOL DECODER

The arrows. Three different notions of "converges," and they are ranked.

$X_n \xrightarrow{q.m.} c$	in mean square (quadratic mean)	$\lim_{n\to\infty} E[(X_n - c)^2] = 0$. The *strongest* of the three and the *easiest to check*, because it splits into bias and variance.
$X_n \xrightarrow{p} c$	in probability	$\lim_{n\to\infty} P(\lvert X_n - c\rvert > \epsilon) = 0$. This is what **consistency** means.
$X_n \xrightarrow{d} Y$	in distribution	$\lim_{n\to\infty} F_{X_n}(r) = F_Y(r)$. The shape of the distribution settles down. This is what the CLT gives you.

The ranking, which you should memorize as one chain:

$$\xrightarrow{q.m.} \implies \xrightarrow{p} \implies \xrightarrow{d}$$

None of the reverse implications hold. **Practical consequence:** to prove consistency ($\xrightarrow{p}$) you almost always prove the stronger mean-square statement instead, because that reduces to two easy limits.

SYMBOL DECODER

The likelihood family. Four related objects that look alike and are constantly confused.

$f(x \mid \theta)$	The pdf or pmf of **one** observation. Given to you on the exam.
$\ell(\theta \mid x_1, \ldots, x_n)$	The **likelihood**: $\prod_{i=1}^{n} f(x_i \mid \theta)$. A product. You rarely work with it directly.
$L(\theta \mid x_1, \ldots, x_n)$	The **log-likelihood**: $\sum_{i=1}^{n} \ln f(x_i \mid \theta)$. A sum. You always work with this one. Logs turn products into sums and sums are differentiable one term at a time.
$S(\theta, x)$	The **score**: $\dfrac{\partial \ln f(x \mid \theta)}{\partial \theta}$. **One observation only. There is no summation sign in the score.**

The single most common error in this entire course: putting a $\sum_{i=1}^{n}$ inside the score function.

The score is the derivative of the log density of a *single* X. The log-likelihood is the sum. They differ by exactly that sum. If your Fisher information comes out with an n in it, you almost certainly made this mistake, because $I(\theta_0)$ is a per-observation quantity and the n appears only later in CRLB $= 1/(nI(\theta_0))$.

SYMBOL DECODER

Information and bounds.

$I(\theta_0)$	**Fisher information for one observation.** Three equivalent formulas, all evaluated at θ_0: $E\left[S^2(\theta_0, X)\right]$, or $\mathrm{Var}[S(\theta_0, X)]$, or $-E[\partial S(\theta_0, X)/\partial\theta]$.
CRLB	$\dfrac{1}{nI(\theta_0)}$. The smallest variance any *unbiased* estimator can have. Note the n: information adds up across observations, so the bound shrinks like $1/n$.
$\mathrm{AVar}(\widehat{\theta})$	Asymptotic variance. If $\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, \Omega)$ then $\mathrm{AVar}(\widehat{\theta}) = \Omega/n$.
$\widehat{\mathrm{AVar}}(\widehat{\theta})$	The *estimated* asymptotic variance: take AVar, which is a function of θ_0, and plug in $\widehat{\theta}$. Justified by the invariance property.

SYMBOL DECODER

Two arrows over the tilde: limiting versus asymptotic distribution. Students lose points here for writing the right thing in the wrong form.

Limiting distribution always has the $\sqrt{n}$ and is centered at zero:

$$\sqrt{n}\,(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, \Omega)$$

Asymptotic (approximate) distribution divides the variance by n and re-centers at θ_0:

$$\widehat{\theta} \overset{a}{\sim} N\left(\theta_0, \tfrac{\Omega}{n}\right)$$

They carry the same content. Move between them by dividing $\sqrt{n}$ across: if $\sqrt{n}(\widehat{\theta} - \theta_0)$ is roughly $N(0, \Omega)$, then $\widehat{\theta} - \theta_0$ is roughly $N(0, \Omega/n)$, so $\widehat{\theta}$ is roughly $N(\theta_0, \Omega/n)$. When a

question says “obtain the limiting distribution,” give the first form. When it says “asymptotic” or “approximate distribution,” give the second. When in doubt on the exam, **write both**. It costs one line and cannot be wrong.

SYMBOL DECODER

Distribution shorthand you will see in the χ^2/t/F question.

$\chi^2_{(p)}$	Chi-squared with p degrees of freedom. Mean p, variance $2p$. Built by adding up p independent squared standard normals.
$t_{(p)}$	Student t with p degrees of freedom. Built as a standard normal over the square root of an independent $\chi^2_{(p)}$ divided by its own p.
$F_{(r_1,r_2)}$	Built as one $\chi^2_{(r_1)}/r_1$ divided by an independent $\chi^2_{(r_2)}/r_2$. Order matters: numerator degrees of freedom first.

SESSION ONE

Three hours to eighty

25 min Block 1 The setting: population, sample, and why $\widehat{\theta}$ is random

35 min Block 2 Sampling distributions: $\overline{X}$ and S^2

30 min Block 3 The Method of Moments machine

40 min Block 4 The Maximum Likelihood machine

35 min Block 5 The asymptotic pipeline: from pdf to $\widehat{\theta} \overset{a}{\sim} N(\cdot, \cdot)$

25 min Block 6 Consistency in two limits

20 min Checkpoint Twelve questions to confirm you are at eighty

By the end of this session you should be able to look at any question on the Sample Final and know within five seconds which machine it wants.

BLOCK 1 The setting: population, sample, and why $\widehat{\theta}$ is random

25 minutes

Everything in the second half of this course is one story, and the story is short. Here it is in five sentences.

There is a population. It has a distribution $f_X(x)$ that you cannot see. You draw a sample $(X_1, \ldots, X_n)$ from it. You compute something from that sample. You use the something to make a claim about the population.

That is it. All the machinery is about doing step four well and knowing how much to trust step five.

SYMBOL DECODER

Random sample (Definition 5.1.1) means exactly two things:

1. $X_1, \ldots, X_n$ are **mutually independent**
2. each X_i has the **same** distribution $f_X(x)$

Independent plus identically distributed, hence "i.i.d." The immediate payoff is the joint density factors:

$$f_{X_1 \cdots X_n}(x_1, \ldots, x_n) = \prod_{i=1}^{n} f_X(x_i)$$

The first equals sign uses independence. Writing the *same* f_X in every factor uses identical distribution. **This product is where the likelihood function comes from.** That is the entire origin story of maximum likelihood.

SYMBOL DECODER

Statistic (Definition 5.2.1): any real-valued function of the sample, $Y = T(X_1, \ldots, X_n)$. Examples: $\overline{X}$, S^2, $\max\{X_i\}$, $X_1 + 3X_7$, the number 8 (a silly but legal statistic).
Sampling distribution: the distribution of that statistic.
The one rule that follows: **a statistic is a function of random variables, so it is itself a random variable**. It has a pdf, a mean, a variance, a cdf. When someone asks "what is $\mathrm{Var}(\widehat{\theta})$?" they are not asking a trick question. They are asking for the variance of a random variable like any other.

WHY THIS IS TRUE

Why is $\overline{X}$ random when the data are just numbers?
Because you could have drawn a different sample. Imagine surveying 1,000 US workers and computing average wage. Now imagine your research assistant had knocked on a different 1,000 doors. She would get a different number. The population mean $E(X)$ did not move; it is a fixed feature of the world. What moved is *which sample you happened to get*. The distribution of $\overline{X}$ describes the spread of answers you would get across all the samples you might have drawn. This is the mental move the whole course rests on. μ is fixed and unknown. $\overline{X}$ is observed and random. Statistics is the art of reasoning from the random thing you see to the fixed thing you

do not.

TRAP

θ_0 is not random. On an exam, never write $\text{Var}(\theta_0)$ or $E(\theta_0)$. The variance of a constant is zero and its expectation is itself. If you find yourself taking the expectation of θ_0, you have confused the truth with your guess. Similarly, $E(\widehat{\theta})$ is a legitimate and important object; $E(\theta_0)$ is just θ_0.

The two approaches, in one line each

Non-parametric: assume nothing about f_X. You can still learn about $E(X)$ and $\text{Var}(X)$. Limited but safe.

Parametric: assume $X \sim f(x \mid \theta_0)$ with the *shape* known and only the *parameters* unknown. Then estimating θ_0 gives you the entire distribution, hence every moment and every probability you might want.

Dr. Flores's summary: the more you assume, the more you get, and the cost is that if the assumption is wrong, everything downstream is wrong. **The whole final exam lives in the parametric world**, except Block 2, which is deliberately distribution-free.

BLOCK 2 Sampling distributions: what we know about $\overline{X}$ and S^2 — 35 minutes

There are two tiers of results here and **knowing which tier you are in is worth real points**. Tier one holds for *any* distribution. Tier two requires normality. Confusing them is a classic strict-grading casualty.

MEMORIZE

TIER ONE (Theorem 5.2.6). Holds for ANY f_X with mean μ and variance σ^2. No normality needed, no distributional assumption at all.

$$\textbf{(a)}\ E(\overline{X}) = \mu \qquad \textbf{(b)}\ \text{Var}(\overline{X}) = \frac{\sigma^2}{n} \qquad \textbf{(c)}\ E(S^2) = \sigma^2$$

In words: the sample mean is centered on the population mean; it gets more precise at rate $1/n$; the $n-1$ version of the sample variance is centered on the population variance. That last one is the entire reason we divide by $n-1$.

MEMORIZE

TIER TWO (Theorem 5.3.1). Requires $(X_1, \ldots, X_n) \overset{iid}{\sim} N(\mu, \sigma^2)$.

$$\textbf{(a)}\ \overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \qquad \textbf{(b)}\ \overline{X} \text{ and } S^2 \text{ are independent} \qquad \textbf{(c)}\ \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

Part (a) upgrades "mean μ, variance σ^2/n" to "it is exactly normal." Parts (b) and (c) are what make the t and F constructions work.

WHY THIS IS TRUE

Why does $\mathrm{Var}(\overline{X}) = \sigma^2/n$ and not σ^2 or $n\sigma^2$?
Watch where the n's go. Variance is not linear; it picks up squares:

$$\mathrm{Var}(\overline{X}) = \mathrm{Var}\left(\frac{1}{n}\sum X_i\right) = \frac{1}{n^2}\mathrm{Var}\left(\sum X_i\right) = \frac{1}{n^2}\sum \mathrm{Var}(X_i) = \frac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n}$$

The $1/n$ out front becomes $1/n^2$ when it comes through the variance. The sum of n variances contributes a factor of n. Net: $n/n^2 = 1/n$.
The middle step, $\mathrm{Var}(\sum X_i) = \sum \mathrm{Var}(X_i)$, **is where independence gets used**, because in general $\mathrm{Var}(\sum X_i) = \sum \mathrm{Var}(X_i) + \sum_{i\neq j}\mathrm{Cov}(X_i, X_j)$, and i.i.d. makes every covariance term zero. **Say this out loud in your proof.** Dr. Flores wants to see the logic, and this is the step he is looking for.
Intuitively: averaging cancels noise. Some draws land high, some low, and they partially offset. More draws, more cancellation, less spread.

WORKED EXAMPLE

Sample Final #1, in full. *Let $X_1, \ldots, X_n$ be an i.i.d. sample with mean μ and variance σ^2. Prove $E(\overline{X}) = \mu$ and $\mathrm{Var}(\overline{X}) = \sigma^2/n$. Be clear about each step.*
This is a gift question and it is worth writing out perfectly, because "be clear about each step" means the steps *are* the grade.
Part 1.

$$\begin{aligned}
E(\overline{X}) &= E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] && \text{definition of } \overline{X}\\
&= \frac{1}{n}E\left[\sum_{i=1}^{n} X_i\right] && \text{expectation is linear, pull out the constant}\\
&= \frac{1}{n}\sum_{i=1}^{n} E(X_i) && \text{expectation of a sum is the sum of expectations}\\
&= \frac{1}{n}\sum_{i=1}^{n} \mu = \frac{1}{n}\cdot n\mu = \mu && \text{identically distributed, so every } E(X_i) = \mu
\end{aligned}$$

Part 2.

$$\begin{aligned}
\mathrm{Var}(\overline{X}) &= \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) && \text{definition} \\
&= \frac{1}{n^2}\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) && \mathrm{Var}(aY) = a^2\mathrm{Var}(Y) \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) && \textbf{by independence: } \mathrm{Cov}(X_i, X_j) = 0,\ i \neq j \\
&= \frac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n} && \text{identically distributed}
\end{aligned}$$

Note the annotations in the right column. That is what "be clear about each step" means. Two places matter most: pulling $1/n$ out of a variance turns it into $1/n^2$, and the third line is where independence enters. Name both.

TRAP

The independence step is the graded step. Every year, students write line three of Part 2 without justification. The line is false without independence. If the problem instead said "correlated sample," you would get $\frac{1}{n^2}\left[\sum \mathrm{Var}(X_i) + \sum_{i\neq j}\mathrm{Cov}(X_i, X_j)\right]$ and the answer would not be σ^2/n. Write the words "by independence."

Two identities you will reuse constantly

The computational form of the sum of squares (Theorem 5.2.4b):

$$(n-1)S^2 = \sum_{i=1}^{n}(X_i - \overline{X})^2 = \sum_{i=1}^{n} X_i^2 - n\overline{X}^2$$

This is how you get from a formula involving $\overline{X}$ inside the sum to one where $\overline{X}$ is outside. It is the key algebraic move in proving $E(S^2) = \sigma^2$.

The variance-to-second-moment flip, used in literally every proof in this course:

$$\mathrm{Var}(Y) = E(Y^2) - [E(Y)]^2 \quad \Longleftrightarrow \quad \boxed{E(Y^2) = \mathrm{Var}(Y) + [E(Y)]^2}$$

The boxed direction is the useful one. Whenever you are stuck with an $E(\text{something}^2)$ you cannot integrate, this converts it into two things you probably already know.

WORKED EXAMPLE

Proof that $E(S^2) = \sigma^2$. You should be able to reproduce this; it or its cousin has appeared.

Start with the computational form:

$$E(S^2) = \frac{1}{n-1} E\left[\sum_{i=1}^{n} X_i^2 - n\overline{X}^2\right] = \frac{1}{n-1}\left[\sum_{i=1}^{n} E(X_i^2) - nE(\overline{X}^2)\right]$$

Now use the flip on both pieces:

$$E(X_i^2) = \text{Var}(X_i) + [E(X_i)]^2 = \sigma^2 + \mu^2, \qquad E(\overline{X}^2) = \text{Var}(\overline{X}) + [E(\overline{X})]^2 = \frac{\sigma^2}{n} + \mu^2$$

Substitute:

$$E(S^2) = \frac{1}{n-1}\left[n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right)\right] = \frac{1}{n-1}\left[n\sigma^2 + n\mu^2 - \sigma^2 - n\mu^2\right] = \frac{(n-1)\sigma^2}{n-1} = \sigma^2$$

The μ^2 terms cancel and the σ^2 terms leave exactly $n-1$ of them. **That is where the $n-1$ in the denominator earns its keep.**

TRAP

$E(S^2) = \sigma^2$ **does NOT imply** $E(S) = \sigma$**.** Dr. Flores raises this in Notes X and again in Notes XIV, so he cares about it.
The reason: expectation does not pass through nonlinear functions. $E[g(Y)] \neq g[E(Y)]$ in general. Square root is concave, so by Jensen's inequality $E(\sqrt{S^2}) < \sqrt{E(S^2)} = \sigma$. The sample standard deviation **underestimates** σ on average.
This is the launching point for the whole asymptotics chapter: unbiasedness is fragile under transformation, whereas consistency is not, because $\text{plim}\, g(\widehat{\theta}) = g(\text{plim}\, \widehat{\theta})$. Hold that contrast; it is Block 6.

Three more facts worth having memorized

(i) Variance of S^2 under normality (PS7 #3, used in PS9 #1 and PS10 #1):

$$\text{Var}(S^2) = \frac{2\sigma^4}{n-1}$$

Derived by applying Var to both sides of $(n-1)S^2/\sigma^2 \sim \chi^2_{(n-1)}$ and using $\text{Var}(\chi^2_{(r)}) = 2r$.
(ii) Two independent samples (PS7 #1):

$$E(\overline{X} - \overline{Y}) = \mu_x - \mu_y, \qquad \text{Var}(\overline{X} - \overline{Y}) = \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}$$

Note the **plus** sign in the variance even though it is a difference of means. Because $\text{Var}(A - B) = \text{Var}A + \text{Var}B - 2\text{Cov}(A, B)$ and the covariance is zero across independent samples.
(iii) The residual is uncorrelated with the mean (PS7 #2):

$$\text{Cov}(X_i - \overline{X},\ \overline{X}) = 0$$

This is the covariance version of Theorem 5.3.1(b) and it is what makes $\overline{X}$ and S^2 independent under normality.

EXAM INTEL FROM DR. FLORES

Sample Final #1 is literally "prove $E(\overline{X}) = \mu$ and $\text{Var}(\overline{X}) = \sigma^2/n$." PS7 #1 and #2 are variations. This family of question is close to guaranteed and it is free points, provided you **annotate every step**. Practice writing the two proofs above from memory until you can do each in under three minutes with the justifications in place.

BLOCK 3 The Method of Moments machine

30 minutes

WHY THIS IS TRUE

The idea in one breath. You want θ_0. You cannot see it. But you know that $E(X)$ is some formula involving θ_0, because the distribution tells you so. And you *do* have a good estimate of $E(X)$, namely $\overline{X}$. So: write down the formula, replace $E(X)$ with $\overline{X}$, and solve for θ. Done. That is the whole method. Everything else is bookkeeping about how many equations you need.

THE RECIPE

Method of Moments, step by step.

1. **Count the parameters.** k parameters means k equations, because k unknowns need k equations.
2. **Write the population moments** as functions of the parameters. Either these are handed to you in the problem, or you compute $E(X) = \int x f(x \mid \theta)\, dx$ yourself.
3. **Set each population moment equal to its sample counterpart:**

$$E(X) = \overline{X} \qquad \text{and, if you need a second equation,} \qquad \text{Var}(X) = \widetilde{S}^2$$

4. **Solve the system algebraically** for the parameters in terms of $\overline{X}$ and $\widetilde{S}^2$.
5. **Put hats on and label:** $\widehat{\theta}_{MOM} = \ldots$

TRAP

Use CENTERED moments for the second equation, and never mix.
There are two legitimate second-equation options and one illegal combination.

$$\text{Centered (use this):} \quad E(X) = \overline{X} \quad \text{and} \quad \text{Var}(X) = \widetilde{S}^2$$

$$\text{Uncentered:} \quad E(X) = \overline{X} \quad \text{and} \quad E(X^2) = \tfrac{1}{n}\textstyle\sum_{i=1}^{n} X_i^2$$

Never write $\frac{1}{n}\sum X_i^2 = \text{Var}(X)$ or $\widetilde{S}^2 = E(X^2)$. Those pair a centered quantity with an uncentered one and they are simply false. Dr. Flores flags this explicitly in Notes XI.
Prefer the centered version, because problems almost always hand you $\text{Var}(X)$ rather than $E(X^2)$.
Also note: the second sample moment here is $\widetilde{S}^2$ with the $\frac{1}{n}$, not S^2 with the $\frac{1}{n-1}$.

WORKED EXAMPLE

One parameter, PS8 #3a. $f(x \mid \theta_0) = \theta_0 x^{\theta_0 - 1}$ *for* $0 < x < 1$. *Given* $E(X) = \theta_0/(\theta_0+1)$. *Find* $\widehat{\theta}_{MOM}$.
One parameter, so one equation. Set $E(X) = \overline{X}$:

$$\frac{\theta}{\theta+1} = \overline{X} \implies \theta = \overline{X}\theta + \overline{X} \implies \theta(1-\overline{X}) = \overline{X} \implies \boxed{\widehat{\theta}_{MOM} = \frac{\overline{X}}{1-\overline{X}}}$$

Two parameters, PS8 #1a. *Gamma*(α_0, β_0) *with* $E(X) = \alpha_0\beta_0$ *and* $\text{Var}(X) = \alpha_0\beta_0^2$. *Find the MOM estimators.*
Two parameters, so two equations:

$$\alpha\beta = \overline{X}, \qquad \alpha\beta^2 = \widetilde{S}^2$$

Divide the second by the first: $\beta = \widetilde{S}^2/\overline{X}$. Substitute back into the first: $\alpha = \overline{X}/\beta = \overline{X}^2/\widetilde{S}^2$. So

$$\boxed{\widehat{\alpha}_{MOM} = \frac{\overline{X}^2}{\widetilde{S}^2}}, \qquad \boxed{\widehat{\beta}_{MOM} = \frac{\widetilde{S}^2}{\overline{X}}}$$

The trick worth stealing: when both equations are products of powers of the parameters, divide one by the other to kill a variable. That works for Gamma, and it is the fastest route.

WORKED EXAMPLE

Two parameters where the second is a standard deviation, PS8 #1b. $f(x \mid \theta_{01}, \theta_{02}) = \frac{1}{2\theta_{02}\sqrt{3}}$ *on* $(\theta_{01} - \theta_{02}\sqrt{3},\ \theta_{01} + \theta_{02}\sqrt{3})$, *with* $E(X) = \theta_{01}$ *and* $\text{Var}(X) = \theta_{02}^2$.

$$\theta_1 = \overline{X} \implies \widehat{\theta}_{1,MOM} = \overline{X}$$

$$\theta_2^2 = \widetilde{S}^2 \implies \widehat{\theta}_{2,MOM} = \sqrt{\widetilde{S}^2} = \widetilde{S}$$

Notice: because θ_{02} is a *standard deviation* rather than a variance, you take a square root at the end. Read carefully which one the parameter is. This distinction reappears in the MLE section and in the delta method question.

WORKED EXAMPLE

When you have to compute $E(X)$ yourself, PS10 #2. $f(x \mid \theta_0) = \frac{1}{2}(1+\theta_0 x)$ *for* $-1 < x < 1$. No mean handed to you, so integrate:

$$E(X) = \int_{-1}^{1} x \cdot \tfrac{1}{2}(1+\theta_0 x)\,dx = \tfrac{1}{2}\int_{-1}^{1}(x+\theta_0 x^2)\,dx = \tfrac{1}{2}\left[\frac{x^2}{2} + \theta_0\frac{x^3}{3}\right]_{-1}^{1} = \tfrac{1}{2}\left[\frac{2\theta_0}{3}\right] = \frac{\theta_0}{3}$$

The $x^2/2$ term vanishes because it is even evaluated symmetrically. Set $\theta_0/3 = \overline{X}$:

$$\boxed{\widehat{\theta}_{MOM} = 3\overline{X}}$$

MEMORIZE

MOM answers you should recognize on sight. These recur across PS8, PS10, and the Sample Final.

Distribution	$E(X)$	$\widehat{\theta}_{MOM}$
Normal(μ_0, σ_0^2)	μ_0, Var $= \sigma_0^2$	$\widehat{\mu} = \overline{X}$, $\widehat{\sigma^2} = \widetilde{S}^2$
Bernoulli(p_0) / Poisson(θ_0)	p_0 / θ_0	$\overline{X}$
Geometric(p_0)	$1/p_0$	$\widehat{p} = 1/\overline{X}$
Exponential, mean form $\frac{1}{\theta_0}e^{-x/\theta_0}$	θ_0	$\overline{X}$
Exponential, rate form $\theta_0 e^{-\theta_0 x}$	$1/\theta_0$	$1/\overline{X}$
Uniform$(0, \theta_0)$	$\theta_0/2$	$2\overline{X}$
$\theta_0 x^{\theta_0 - 1}$ on $(0,1)$	$\theta_0/(\theta_0+1)$	$\overline{X}/(1-\overline{X})$
Gamma(α_0, β_0)	$\alpha_0\beta_0$	$\widehat{\alpha} = \overline{X}^2/\widetilde{S}^2$, $\widehat{\beta} = \widetilde{S}^2/\overline{X}$

TRAP

Two known weaknesses of MOM, either of which could be a one-line exam part.

1. **It can produce estimates outside the parameter space.** For a Binomial(n_0, p_0), MOM can return $\widehat{p} < 0$ when $\widetilde{S}^2 > \overline{X}$, which is nonsense since $p \in [0,1]$.
2. **It is not always efficient.** MLE often does better asymptotically.

Its compensating virtue: MOM almost always has a closed form. For the Gamma, the MLE has *no* closed form and must be computed numerically, while MOM falls right out. That contrast is exactly why PS8 #1a exists.

BLOCK 4 The Maximum Likelihood machine

40 minutes

WHY THIS IS TRUE

The idea in one breath. You got the sample you got. For each candidate value of θ, ask: how probable was this exact sample, if θ were the truth? Then pick the θ that makes your data look least surprising.
Concretely: suppose you flip a coin ten times and see eight heads. If $p = 0.1$, that outcome is wildly improbable. If $p = 0.8$, it is the single most probable outcome. MLE says pick 0.8. That is the entire intuition, and it is correct: for Bernoulli, $\widehat{p}_{MLE} = \overline{X}$.

SYMBOL DECODER

From product to sum, which is the only technical idea here.
The probability of the whole sample is a *product*, because the observations are independent:

$$\ell(\theta \mid x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i \mid \theta)$$

Products are miserable to differentiate. So take logs, which turns the product into a sum:

$$L(\theta \mid x_1, \ldots, x_n) = \ln \ell = \sum_{i=1}^{n} \ln f(x_i \mid \theta)$$

Logarithm is strictly increasing, so **whatever maximizes ℓ also maximizes L.** You lose nothing and gain a sum you can differentiate term by term.

THE RECIPE

Maximum Likelihood, step by step. Learn this as muscle memory.

1. **Take the log of the density for ONE observation:** $\ln f(x \mid \theta)$. Expand it fully. Split products into sums of logs, bring exponents down as multipliers, and **drop nothing**, though you may note which terms have no θ in them.
2. **Sum over i:** $L(\theta) = \sum_{i=1}^{n} \ln f(x_i \mid \theta)$. Terms free of θ become constants; terms with x_i in them become $\sum x_i$ or $\sum \ln x_i$ or $\sum x_i^2$ as appropriate.
3. **Differentiate with respect to θ and set to zero** (the first-order condition).
4. **Solve for θ.** Put a hat on it and write X capitalized in the final answer, to stress the estimator is random.

You do not need to check second-order conditions. Dr. Flores says so explicitly in the PS8 answer key: "in the test you can feel free not to check the second order conditions."

TRAP

Differentiate with respect to the actual parameter. If the parameter is σ_0^2, differentiate with respect to σ^2 treated as a single symbol, **not** with respect to σ. Dr. Flores calls this out in the PS8 key. A clean way to avoid the error: substitute $v = \sigma^2$, do all the calculus in v, then translate back at the end.

WORKED EXAMPLE

Poisson (PS8 #2a). He said to expect something like this. $f(x \mid \theta_0) = \dfrac{\theta_0^x e^{-\theta_0}}{x!}$ *for* $x = 0, 1, 2, \ldots$

Step 1. $\ln f(x \mid \theta) = x \ln \theta - \theta - \ln(x!)$

Step 2. $L(\theta) = \sum_{i=1}^{n} X_i \ln \theta - n\theta - \sum_{i=1}^{n} \ln(X_i!)$

Step 3. $\dfrac{\partial L}{\partial \theta} = \dfrac{1}{\theta} \sum_{i=1}^{n} X_i - n = 0$

Step 4. $\sum X_i = n\theta \implies \boxed{\widehat{\theta}_{MLE} = \dfrac{1}{n} \sum_{i=1}^{n} X_i = \overline{X}}$

The $\ln(X_i!)$ term has no θ, so it dies at step 3. **Do not panic when you see a term you cannot simplify; check whether it even survives the derivative.**

WORKED EXAMPLE

Exponential in mean form (PS8 #2b). $f(x \mid \theta_0) = \dfrac{1}{\theta_0} e^{-x/\theta_0}$ *for* $x > 0$.

Step 1. $\ln f = -\ln \theta - \dfrac{x}{\theta}$

Step 2. $L(\theta) = -n \ln \theta - \dfrac{1}{\theta} \sum_{i=1}^{n} X_i$

Step 3. $\dfrac{\partial L}{\partial \theta} = -\dfrac{n}{\theta} + \dfrac{1}{\theta^2} \sum X_i = 0$

Step 4. Multiply through by θ^2: $-n\theta + \sum X_i = 0 \implies \boxed{\widehat{\theta}_{MLE} = \overline{X}}$

Compare with the rate form $f(x \mid \theta_0) = \theta_0 e^{-\theta_0 x}$ (PS10 #5), which is the same distribution written with θ meaning $1/\text{mean}$:

$$\ln f = \ln \theta - \theta x, \quad L = n \ln \theta - \theta \sum X_i, \quad \frac{\partial L}{\partial \theta} = \frac{n}{\theta} - \sum X_i = 0 \implies \boxed{\widehat{\theta}_{MLE} = \frac{1}{\overline{X}}}$$

Same distribution, reciprocal answers. Always read which parameterization you are handed. If $E(X) = \theta$ the answer is $\overline{X}$; if $E(X) = 1/\theta$ the answer is $1/\overline{X}$.

WORKED EXAMPLE

Normal with mean zero, unknown variance (PS8 #4a). $f(x \mid \sigma_0^2) = \dfrac{1}{\sqrt{2\pi\sigma_0^2}} e^{-x^2/(2\sigma_0^2)}$

Step 1. $\ln f = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma^2) - \dfrac{x^2}{2\sigma^2}$

Step 2. $L = -\dfrac{n}{2} \ln(2\pi) - \dfrac{n}{2} \ln(\sigma^2) - \dfrac{1}{2\sigma^2} \sum X_i^2$

Step 3. Differentiating with respect to σ^2 (as one symbol): $\dfrac{\partial L}{\partial \sigma^2} = -\dfrac{n}{2\sigma^2} + \dfrac{1}{2\sigma^4} \sum X_i^2 = 0$

Step 4. $\dfrac{1}{2\sigma^4} \sum X_i^2 = \dfrac{n}{2\sigma^2} \implies \boxed{\widehat{\sigma^2}_{MLE} = \dfrac{1}{n} \sum_{i=1}^{n} X_i^2}$

Note that with μ known to be zero, you do *not* subtract $\overline{X}$. With μ unknown, the MLE would be $\widetilde{S}^2 = \frac{1}{n}\sum(X_i - \overline{X})^2$ instead.

WORKED EXAMPLE

A case where MLE and MOM disagree (PS8 #3a). $f(x \mid \theta_0) = \theta_0 x^{\theta_0 - 1}$ *on* $(0,1)$.
Step 1. $\ln f = \ln\theta + (\theta - 1)\ln x$
Step 2. $L = n\ln\theta + (\theta - 1)\sum_{i=1}^{n} \ln X_i$
Step 3. $\dfrac{\partial L}{\partial \theta} = \dfrac{n}{\theta} + \sum_{i=1}^{n} \ln X_i = 0$
Step 4. $\boxed{\widehat{\theta}_{MLE} = \dfrac{-n}{\sum_{i=1}^{n} \ln X_i}}$
The minus sign is correct and not a mistake: for $x \in (0,1)$, $\ln x < 0$, so the denominator is negative and $\widehat{\theta} > 0$. Recall from Block 3 that $\widehat{\theta}_{MOM} = \overline{X}/(1 - \overline{X})$ here. **They genuinely differ.** MOM and MLE coincide sometimes (Normal, Poisson, Bernoulli, exponential in mean form) but not always.

The invariance property, which is free points

MEMORIZE

Invariance Property of the MLE (Theorem 7.2.10). If $\widehat{\theta}_{MLE}$ is the MLE of θ_0, then for any function g, the MLE of $\gamma_0 = g(\theta_0)$ is simply

$$\widehat{\gamma}_{MLE} = g(\widehat{\theta}_{MLE})$$

You just plug in. No new maximization, no new derivatives, no work whatsoever.

WORKED EXAMPLE

Three instances that have all appeared:

- MLE of σ_0^2 is $\widetilde{S}^2$, so the MLE of $\sigma_0 = \sqrt{\sigma_0^2}$ is $\widetilde{S} = \sqrt{\widetilde{S}^2}$. (PS10 #3a)
- Bernoulli MLE of p_0 is $\overline{X}$, so the MLE of $\gamma_0 = 1/p_0$ is $1/\overline{X}$. (SF #4g)
- Poisson MLE of θ_0 is $\overline{X}$, so the MLE of $\gamma_0 = \Pr(X = 0) = e^{-\theta_0}$ is $e^{-\overline{X}}$. (PS10 #6b)

This is also how you estimate an asymptotic variance, which is Block 5. AVar is a function of θ_0; invariance lets you replace θ_0 by $\widehat{\theta}$ and call the result $\widehat{\text{AVar}}$.

TRAP

Invariance holds for MLE but NOT for unbiasedness. $E[g(\widehat{\theta})] \neq g[E(\widehat{\theta})]$. The MLE of σ is $\widetilde{S}$, but $\widetilde{S}$ is *not* unbiased for σ even though $\widetilde{S}^2$ is nearly unbiased for σ^2. Invariance is a statement about *maximizers*, not about *expectations*. Consistency, on the other hand, does survive

continuous transformation, which is Block 6.

TRAP

When calculus fails. Two cases have appeared and either could be a short exam part.

- **Support depends on θ.** Uniform$(0,\theta_0)$: writing $f(x_i \mid \theta_0) = \frac{1}{\theta_0}\mathbf{1}\{x_i \le \theta_0\}$ gives $L = -n\ln\theta + \sum \ln \mathbf{1}\{X_i \le \theta\}$. Calculus cannot handle the indicator. Reason it out instead: you need $\theta \ge \max\{X_i\}$ or the likelihood is $-\infty$, and $-n\ln\theta$ is decreasing in θ, so take θ as small as legally allowed. Answer: $\widehat{\theta}_{MLE} = \max\{X_i\}$.
- **Non-differentiable density.** Laplace, $f = \frac{1}{2}e^{-|x-\theta_0|}$: the absolute value is not differentiable, and maximizing L is the same as minimizing $\frac{1}{n}\sum|X_i - \theta|$. Answer: the sample median. Dr. Flores explicitly said not to derive this and only wanted the log-likelihood, so **if you see an absolute value, write the log-likelihood and state that the MLE is the sample median**.

BLOCK 5 The asymptotic pipeline: from a pdf to $\widehat{\theta} \overset{a}{\sim} N(\cdot,\cdot)$ 35 minutes

EXAM INTEL FROM DR. FLORES

This block is the highest-value forty minutes in the entire guide. Dr. Flores listed "find the asymptotic distribution of the MLE" as a for-sure question, and the *same pipeline* also answers the CRLB question and the UMVUE question. **One machine, three exam questions.** Learn it cold.

THE RECIPE

THE PIPELINE. Given a density $f(x \mid \theta_0)$, this produces everything.

STEP 0. Log the density (one observation).

$$\ln f(x \mid \theta)$$

STEP 1. Score: differentiate once, no summation.

$$S(\theta, X) = \frac{\partial \ln f(X \mid \theta)}{\partial \theta}$$

STEP 2. Fisher information: pick whichever of the three is easiest.

$$I(\theta_0) = \underbrace{-E\left[\frac{\partial S(\theta_0, X)}{\partial \theta}\right]}_{\text{usually easiest}} = \text{Var}[S(\theta_0, X)] = E\left[S^2(\theta_0, X)\right]$$

STEP 3. Read off both answers at once.

$$\text{CRLB} = \frac{1}{nI(\theta_0)} \qquad \text{and} \qquad \sqrt{n}\left(\widehat{\theta}_{MLE} - \theta_0\right) \xrightarrow{d} N\left(0,\ [I(\theta_0)]^{-1}\right)$$

STEP 4. Convert to the asymptotic form and, if asked, estimate it.

$$\widehat{\theta}_{MLE} \overset{a}{\sim} N\left(\theta_0, \frac{1}{nI(\theta_0)}\right), \qquad \widehat{\mathrm{AVar}}(\widehat{\theta}_{MLE}) = \left.\frac{1}{nI(\theta)}\right|_{\theta=\widehat{\theta}_{MLE}}$$

MEMORIZE

The punchline that ties it together: the asymptotic variance of the MLE *equals* the CRLB. That is what "asymptotically efficient" means. So the moment you have computed $I(\theta_0)$, you have simultaneously answered: what is the CRLB, what is the limiting distribution of the MLE, what is its asymptotic distribution, and how would you estimate its variance. **Four exam parts, one calculation.**

SYMBOL DECODER

Which of the three information formulas should I use?
Dr. Flores's own handwritten note on the Sample Final answer key works one problem all three ways and remarks that which is easiest depends on the problem. Practical guidance:

- $-E[\partial S/\partial\theta]$ **is usually easiest** when differentiating the score again produces something with no X in it, or with only X to the first power. Then the expectation is trivial. This was the recommended route in PS10 #4 and #5.
- $\mathrm{Var}[S(\theta_0, X)]$ **is easiest** when the score is linear in X, because then $\mathrm{Var}[a + bX] = b^2\mathrm{Var}(X)$ and you just quote $\mathrm{Var}(X)$. This was the recommended route for the Bernoulli in the Sample Final key, and for Poisson and exponential in PS9 #2.
- $E[S^2]$ **is the fallback**, and it is the definition you must use when the regularity conditions fail (see the trap in Block 12).

Rule of thumb: look at the score. If it is linear in X, use $\mathrm{Var}[S]$. Otherwise differentiate again.

WORKED EXAMPLE

Full pipeline, exponential rate form (PS10 #5). $f(x \mid \theta_0) = \theta_0 e^{-\theta_0 x}$, $x > 0$. *Recall* $E(X) = 1/\theta_0$ *and* $\mathrm{Var}(X) = 1/\theta_0^2$.

Step 0. $\ln f = \ln\theta - \theta x$

Step 1. $S(\theta, X) = \dfrac{1}{\theta} - X$ (one observation, no sum)

Step 2. The score is linear in X, so both routes are easy. Differentiating again: $\dfrac{\partial S}{\partial\theta} = -\dfrac{1}{\theta^2}$, which has no X, so

$$I(\theta_0) = -E\left[-\frac{1}{\theta_0^2}\right] = \frac{1}{\theta_0^2}$$

Check via the other route: $\mathrm{Var}[S] = \mathrm{Var}\left[\frac{1}{\theta_0} - X\right] = \mathrm{Var}(X) = \frac{1}{\theta_0^2}$. Same answer.

Step 3. $\text{CRLB} = \dfrac{1}{n \cdot 1/\theta_0^2} = \dfrac{\theta_0^2}{n}$ and, using $\widehat{\theta}_{MLE} = 1/\overline{X}$ from Block 4,

$$\sqrt{n}\,(\widehat{\theta}_{MLE} - \theta_0) \xrightarrow{d} N(0, \theta_0^2)$$

Step 4. $\widehat{\theta}_{MLE} \overset{a}{\sim} N\left(\theta_0, \dfrac{\theta_0^2}{n}\right)$ and $\widehat{\text{AVar}} = \dfrac{\widehat{\theta}_{MLE}^2}{n} = \dfrac{1}{n\overline{X}^2}$

WORKED EXAMPLE

Full pipeline, PS10 #4. $f(x \mid \theta_0) = \theta_0 x^{\theta_0 - 1}$ *on* $(0,1)$. *Find an estimator that is consistent, asymptotically normal and asymptotically efficient, and give its asymptotic distribution.*
The phrase "consistent, asymptotically normal, asymptotically efficient" is a coded instruction: it is quoting the MLE theorem word for word. **The answer is "use the MLE."**
From Block 4, $\widehat{\theta}_{MLE} = -n/\sum \ln X_i$.
Step 0–1. $\ln f = \ln\theta + (\theta - 1)\ln x$, so $S(\theta, X) = \dfrac{1}{\theta} + \ln X$.
Step 2. Here $\text{Var}[S]$ would require $\text{Var}(\ln X)$, which is extra work. Differentiate again instead: $\partial S/\partial\theta = -1/\theta^2$, so $I(\theta_0) = 1/\theta_0^2$.
Step 3–4. $\sqrt{n}(\widehat{\theta}_{MLE} - \theta_0) \xrightarrow{d} N(0, \theta_0^2)$ and $\widehat{\theta}_{MLE} \overset{a}{\sim} N(\theta_0, \theta_0^2/n)$, and by invariance

$$\widehat{\text{AVar}}(\widehat{\theta}_{MLE}) = \frac{\widehat{\theta}_{MLE}^2}{n} = \frac{n}{\left[\sum_{i=1}^n \ln X_i\right]^2}$$

Attaching the delta method to the end of the pipeline

MEMORIZE

Delta Method. If $\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, \Lambda)$ and g is continuously differentiable with $g'(\theta_0) \neq 0$, then

$$\sqrt{n}\left[g(\widehat{\theta}) - g(\theta_0)\right] \xrightarrow{d} N\left(0,\ [g'(\theta_0)]^2\,\Lambda\right)$$

In words: the new variance is the old variance times the squared derivative. That is the whole content.

THE RECIPE

When a question says "estimate $\gamma_0 = g(\theta_0)$ and give its asymptotic distribution," do exactly this:

1. **Invariance** gives you the estimator instantly: $\widehat{\gamma} = g(\widehat{\theta})$.
2. **Compute** $g'(\theta)$, then square it, then evaluate at θ_0.
3. **Multiply** by the Λ from the pipeline (which for an MLE is $[I(\theta_0)]^{-1}$).
4. **Write both forms:** $\sqrt{n}(\widehat{\gamma} - \gamma_0) \xrightarrow{d} N(0, [g'(\theta_0)]^2\Lambda)$ and $\widehat{\gamma} \overset{a}{\sim} N(\gamma_0, [g'(\theta_0)]^2\Lambda/n)$.

WORKED EXAMPLE

Delta method, Sample Final #4g. *Bernoulli*(p_0). *Estimate* $\gamma_0 = 1/p_0$ *and derive its asymptotic distribution.*

1. $\widehat{p}_{MLE} = \overline{X}$, so by invariance $\widehat{\gamma}_{MLE} = 1/\overline{X}$.

2. $g(p) = 1/p$, so $g'(p) = -1/p^2$ and $[g'(p_0)]^2 = 1/p_0^4$.

3. From the pipeline, $\Lambda = p_0(1-p_0)$. Multiply:

$$\frac{1}{p_0^4} \cdot p_0(1-p_0) = \frac{1-p_0}{p_0^3}$$

4. $\sqrt{n}(\widehat{\gamma}_{MLE} - \gamma_0) \xrightarrow{d} N\left(0, \frac{1-p_0}{p_0^3}\right)$ and $\widehat{\gamma}_{MLE} \overset{a}{\sim} N\left(\frac{1}{p_0}, \frac{1-p_0}{np_0^3}\right)$

WORKED EXAMPLE

Delta method, PS10 #3b. *Given* $\sqrt{n}(\widehat{\sigma^2}_{MLE} - \sigma_0^2) \xrightarrow{d} N(0, 2\sigma_0^4)$, *find the asymptotic distribution of* $\widehat{\sigma}_{MLE}$.

1. By invariance $\widehat{\sigma}_{MLE} = (\widehat{\sigma^2}_{MLE})^{1/2}$.

2. Treat $v = \sigma^2$ as the argument. $g(v) = \sqrt{v}$, so $g'(v) = \frac{1}{2}v^{-1/2}$ and $[g'(\sigma_0^2)]^2 = \frac{1}{4\sigma_0^2}$.

3. $\frac{1}{4\sigma_0^2} \cdot 2\sigma_0^4 = \frac{\sigma_0^2}{2}$

4. $\sqrt{n}(\widehat{\sigma}_{MLE} - \sigma_0) \xrightarrow{d} N(0, \frac{1}{2}\sigma_0^2)$ and $\widehat{\sigma}_{MLE} \overset{a}{\sim} N\left(\sigma_0, \frac{\sigma_0^2}{2n}\right)$

Note the substitution trick in step 2. Renaming σ^2 as v prevents the single most common slip here, which is differentiating with respect to σ instead of σ^2.

WORKED EXAMPLE

Delta method, PS10 #6b. *Poisson*(θ_0). *Estimate* $\gamma_0 = \Pr(X = 0) = e^{-\theta_0}$.

$\widehat{\theta}_{MLE} = \overline{X}$ and $I(\theta_0) = 1/\theta_0$ so $\Lambda = \theta_0$. By invariance $\widehat{\gamma} = e^{-\overline{X}}$. Then $g(\theta) = e^{-\theta}$, $g'(\theta) = -e^{-\theta}$, $[g']^2 = e^{-2\theta_0}$. So

$$\sqrt{n}(\widehat{\gamma} - \gamma_0) \xrightarrow{d} N(0,\ e^{-2\theta_0}\theta_0) \qquad \text{and} \qquad \widehat{\gamma} \overset{a}{\sim} N\left(e^{-\theta_0},\ \frac{e^{-2\theta_0}\theta_0}{n}\right)$$

TRAP

Square the derivative. Do not forget, and do not square twice. The variance multiplier is $[g'(\theta_0)]^2$, evaluated at the true parameter, and it enters the *variance* slot, not the standard deviation slot. Also: the delta method output is centered at $g(\theta_0)$, not at θ_0. Write $\sqrt{n}[g(\widehat{\theta}) - g(\theta_0)]$, never $\sqrt{n}[g(\widehat{\theta}) - \theta_0]$.

BLOCK 6 Consistency in two limits

25 minutes

SYMBOL DECODER

What consistency actually says. $\widehat{\theta}$ is consistent for θ_0 if $\widehat{\theta} \xrightarrow{p} \theta_0$, meaning

$$\lim_{n\to\infty} P\left(|\widehat{\theta} - \theta_0| < \epsilon\right) = 1 \quad \text{for every } \epsilon > 0$$

Plain English: pick any tolerance you like, however tight. With enough data, the probability that your estimator lands inside that tolerance goes to one. **Consistency is the minimum standard of decency for an estimator.** An inconsistent estimator does not improve with more data, which makes it close to useless.

THE RECIPE

How to prove consistency on an exam. This is a two-line ritual.

1. Compute $E(\widehat{\theta})$, hence $\text{Bias}(\widehat{\theta}) = E(\widehat{\theta}) - \theta_0$. Show $\lim_{n\to\infty} \text{Bias}(\widehat{\theta}) = 0$.
2. Compute $\text{Var}(\widehat{\theta})$. Show $\lim_{n\to\infty} \text{Var}(\widehat{\theta}) = 0$.
3. Conclude: *therefore* $\widehat{\theta} \xrightarrow{q.m.} \theta_0$, *and since mean-square convergence implies convergence in probability,* $\widehat{\theta} \xrightarrow{p} \theta_0$, *so* $\widehat{\theta}$ *is consistent for* θ_0.

Write that third sentence. It is the step that earns the point, because it names the logical bridge from the two limits to the definition.

WHY THIS IS TRUE

Why do those two limits suffice? Because

$$E\left[(\widehat{\theta} - \theta_0)^2\right] = \text{Var}(\widehat{\theta}) + \text{Bias}^2(\widehat{\theta})$$

which is just the MSE decomposition. If both pieces go to zero, the whole MSE goes to zero, which is exactly the definition of $\xrightarrow{q.m.}$. And $\xrightarrow{q.m.} \Rightarrow \xrightarrow{p}$. So checking two easy limits establishes the harder-sounding probabilistic statement. **This is why the course teaches mean-square convergence at all: it is a convenient ladder to consistency.**
Careful in the other direction: the converse fails. An estimator can be consistent without its variance going to zero, so failing this test does not prove inconsistency. But on this exam, every consistency question is designed to pass it.

WORKED EXAMPLE

PS10 #1. $(X_1, \ldots, X_n) \overset{iid}{\sim} N(\mu_0, \sigma_0^2)$. *Show* S^2 *is consistent for* σ_0^2.
Bias. By Theorem 5.2.6(c), $E(S^2) = \sigma_0^2$ for any distribution, so $\text{Bias}(S^2) = 0$ and trivially $\lim_{n\to\infty} \text{Bias}(S^2) = 0$.

Variance. Under normality, $\text{Var}(S^2) = \dfrac{2\sigma_0^4}{n-1}$, so

$$\lim_{n\to\infty} \text{Var}(S^2) = \lim_{n\to\infty} \frac{2\sigma_0^4}{n-1} = 0$$

Conclude. $S^2 \xrightarrow{q.m.} \sigma_0^2$, which implies $S^2 \xrightarrow{p} \sigma_0^2$. Therefore S^2 is consistent for σ_0^2. ■

WORKED EXAMPLE

PS10 #2, the full version. This is the pattern he called a for-sure question. $f(x \mid \theta_0) = \frac{1}{2}(1+\theta_0 x)$ *on* $(-1,1)$. *Find a consistent estimator.*
Find one first. From Block 3, $E(X) = \theta_0/3$, so $\widehat{\theta}_{MOM} = 3\overline{X}$.
Bias. $E(\widehat{\theta}) = 3E(\overline{X}) = 3E(X) = 3 \cdot \dfrac{\theta_0}{3} = \theta_0$. Unbiased, so $\lim \text{Bias} = 0$.
Variance. We need $\text{Var}(X)$, so compute the second moment:

$$E(X^2) = \int_{-1}^{1} x^2 \cdot \tfrac{1}{2}(1+\theta_0 x)\,dx = \tfrac{1}{2}\left[\frac{x^3}{3} + \theta_0 \frac{x^4}{4}\right]_{-1}^{1} = \tfrac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$$

Then $\text{Var}(X) = E(X^2) - [E(X)]^2 = \dfrac{1}{3} - \dfrac{\theta_0^2}{9}$, so

$$\text{Var}(\widehat{\theta}) = \text{Var}(3\overline{X}) = 9\text{Var}(\overline{X}) = \frac{9\text{Var}(X)}{n} = \frac{9}{n}\left[\frac{1}{3} - \frac{\theta_0^2}{9}\right] = \frac{1}{n}\left[3 - \theta_0^2\right]$$

$$\lim_{n\to\infty} \text{Var}(\widehat{\theta}) = \lim_{n\to\infty} \frac{3-\theta_0^2}{n} = 0$$

Conclude. Both limits are zero, so $\widehat{\theta}_{MOM} = 3\overline{X} \xrightarrow{p} \theta_0$ and is consistent. ■
Structure to memorize: find estimator, take expectation, take variance, take two limits, say the magic sentence. Almost every consistency question on this exam is that, with different integrals in the middle.

MEMORIZE

Two shortcuts that save real time.
(1) The Weak Law of Large Numbers. For any i.i.d. sample with finite mean and variance, $\overline{X} \xrightarrow{p} \mu$. So if your estimator *is* the sample mean, or a constant times it, you can invoke WLLN in one line rather than computing limits. (The proof is one application of Chebyshev's inequality: $P(|\overline{X} - \mu| > \epsilon) \le \text{Var}(\overline{X})/\epsilon^2 = \sigma^2/(n\epsilon^2) \to 0$.)
(2) Continuous mapping. If $\widehat{\theta} \xrightarrow{p} \theta_0$ and g is continuous, then $g(\widehat{\theta}) \xrightarrow{p} g(\theta_0)$. Equivalently, $\text{plim}\, g(\widehat{\theta}) = g(\text{plim}\, \widehat{\theta})$.
Shortcut (2) is the workhorse. To show $\widetilde{S}$ is consistent for σ_0: first show $\widetilde{S}^2 \xrightarrow{p} \sigma_0^2$ by the two limits, then apply continuous mapping with $g(u) = \sqrt{u}$. Done. **Contrast this with unbiasedness, which does not survive g at all.** That contrast is exactly why asymptotics exists

> as a chapter.

TRAP

Do not confuse the two properties of $\widetilde{S}^2$. It is *biased* in finite samples, since $E(\widetilde{S}^2) = \frac{n-1}{n}\sigma_0^2 \neq \sigma_0^2$. But it is *asymptotically unbiased*, because $\frac{n-1}{n} \to 1$, and combined with $\text{Var}(\widetilde{S}^2) = \frac{2(n-1)}{n^2}\sigma_0^4 \to 0$ it is consistent. **Biased and consistent are perfectly compatible.** A question can legitimately ask you to say "biased, but consistent," and both halves need saying.

CHECKPOINT Twelve questions. Answer without looking back. 20 minutes

If you can do ten of these cold, you are at eighty percent. Answers are on the next page.

1. State the two conditions that make $(X_1, \ldots, X_n)$ a random sample, and write the resulting joint density.
2. Which of $E(\overline{X}) = \mu$, $\text{Var}(\overline{X}) = \sigma^2/n$, $E(S^2) = \sigma^2$, and $\overline{X} \sim N(\mu, \sigma^2/n)$ require normality?
3. Write S^2 and $\widetilde{S}^2$ and state which one is unbiased and which one is the MLE of σ^2.
4. Give the two equations you would set up to find the MOM estimators of a two-parameter distribution, using centered moments.
5. List the four steps of the MLE recipe.
6. Define the score function. How many summation signs does it contain?
7. Write the three formulas for Fisher information $I(\theta_0)$.
8. Write the CRLB in terms of $I(\theta_0)$ and n.
9. Write the limiting distribution of the MLE and then convert it to the asymptotic distribution.
10. State the delta method, including where the squared derivative goes.
11. Give the two limits you check to prove consistency, and state the implication chain that connects them to $\widehat{\theta} \xrightarrow{p} \theta_0$.
12. For $f(x \mid \theta_0) = \theta_0 e^{-\theta_0 x}$ on $x > 0$: find the MLE, the score, $I(\theta_0)$, the CRLB, and the asymptotic distribution of $\widehat{\theta}_{MLE}$.

Checkpoint answers

1. Mutually independent, and identically distributed. Joint density $\prod_{i=1}^{n} f_X(x_i)$.
2. Only the last one. The first three are Theorem 5.2.6 and hold for any distribution with finite mean and variance.
3. $S^2 = \frac{1}{n-1}\sum(X_i - \overline{X})^2$ is unbiased. $\widetilde{S}^2 = \frac{1}{n}\sum(X_i - \overline{X})^2$ is the MOM and MLE estimator and is biased downward.
4. $E(X) = \overline{X}$ and $\text{Var}(X) = \widetilde{S}^2$. Never mix centered and uncentered.
5. Log the density for one observation; sum over i; differentiate with respect to θ and set to zero; solve.
6. $S(\theta, X) = \partial \ln f(X \mid \theta)/\partial\theta$. **Zero summation signs.** It is a one-observation object.
7. $I(\theta_0) = E[S^2(\theta_0, X)] = \text{Var}[S(\theta_0, X)] = -E[\partial S(\theta_0, X)/\partial\theta]$, all evaluated at θ_0.
8. $\text{CRLB} = \dfrac{1}{nI(\theta_0)}$.
9. $\sqrt{n}(\widehat{\theta}_{MLE} - \theta_0) \xrightarrow{d} N(0, [I(\theta_0)]^{-1})$, therefore $\widehat{\theta}_{MLE} \overset{a}{\sim} N\left(\theta_0, \dfrac{1}{nI(\theta_0)}\right)$.
10. If $\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, \Lambda)$ then $\sqrt{n}[g(\widehat{\theta}) - g(\theta_0)] \xrightarrow{d} N(0, [g'(\theta_0)]^2\Lambda)$. The squared derivative multiplies the *variance*.
11. $\lim_{n\to\infty} \text{Bias}(\widehat{\theta}) = 0$ and $\lim_{n\to\infty} \text{Var}(\widehat{\theta}) = 0$. These give $\widehat{\theta} \xrightarrow{q.m.} \theta_0$, and $\xrightarrow{q.m.} \Rightarrow \xrightarrow{p}$, so $\widehat{\theta}$ is consistent.
12. $\widehat{\theta}_{MLE} = 1/\overline{X}$; $S(\theta, X) = 1/\theta - X$; $I(\theta_0) = 1/\theta_0^2$; $\text{CRLB} = \theta_0^2/n$; $\widehat{\theta}_{MLE} \overset{a}{\sim} N(\theta_0, \theta_0^2/n)$.

MEMORIZE

Scored eight or fewer? The gaps are almost always in items 6 through 10, which are the pipeline. Reread Block 5 and rewrite the pipeline from memory on a blank sheet three times. It is the single highest-yield thing you can do with twenty minutes.

SESSION TWO

Five hours to one hundred

40 min Block 7 The χ^2, t, and F construction machine

30 min Block 8 The score identity and the information matrix identity

45 min Block 9 The CRLB three ways, and the UMVUE decision

30 min Block 10 Comparing estimators by mean squared error

40 min Block 11 Convergence, WLLN, CLT, Slutsky, delta method

25 min Block 12 Every trap, and the strict-grading checklist

50 min Block 13 The real Sample Final, fully solved

60 min Block 14 Mock final: ten questions, thirteen parts, with solutions

Session 1 taught you which machine to reach for. Session 2 is about never dropping a step, never mislabeling a limiting distribution, and recognizing the three places where the standard theorems quietly stop applying.

BLOCK 7 The χ^2, t, and F construction machine

40 minutes

EXAM INTEL FROM DR. FLORES

Dr. Flores singled this out: "For sure, you can expect in the exam a question like #5, where you need to construct a given distribution (Chi-squared, t or F) based on normally distributed random variables." Sample Final #5 and PS7 #5 are the templates. **This is the closest thing to a guaranteed question on the exam**, and it is entirely mechanical once you see the structure.

The four building blocks

Everything in this block is assembled from four facts. Learn them as a ladder, because each rung is built from the one below.

MEMORIZE

RUNG 1. Standardize. If $X \sim N(\mu, \sigma^2)$ then

$$\frac{X-\mu}{\sigma} \sim N(0,1)$$

Divide by the standard *deviation*, not the variance. If a problem writes $X \sim N(\mu_x, \sigma_x^2)$, the denominator is σ_x.

RUNG 2. Square a standard normal to get one degree of freedom.

$$Z \sim N(0,1) \implies Z^2 \sim \chi^2_{(1)}$$

RUNG 3. Add independent chi-squares and the degrees of freedom add.

$$Y_i \sim \chi^2_{(p_i)} \text{ independent} \implies \sum_{i=1}^{m} Y_i \sim \chi^2_{(\sum p_i)}$$

Combining rungs 2 and 3: m independent standard normals, squared and summed, give $\chi^2_{(m)}$.

RUNG 4a. The t recipe (Result 1). If $U \sim N(0,1)$, $V \sim \chi^2_{(p)}$, and U and V are **independent**, then

$$\frac{U}{\sqrt{V/p}} \sim t_{(p)}$$

RUNG 4b. The F recipe (Result 2). If $U \sim \chi^2_{(r_1)}$, $V \sim \chi^2_{(r_2)}$, **independent**, then

$$\frac{U/r_1}{V/r_2} \sim F_{(r_1, r_2)}$$

SYMBOL DECODER

Read the recipes as shapes, not formulas.

$$t_{(p)} = \frac{\text{normal}}{\sqrt{\dfrac{\text{chi-squared}}{\text{its own df}}}} \qquad F_{(r_1,r_2)} = \frac{\dfrac{\text{chi-squared}}{\text{its own df}}}{\dfrac{\text{another chi-squared}}{\text{its own df}}}$$

Two things to burn in:

- **Every chi-squared gets divided by its own degrees of freedom.** A $\chi^2_{(2)}$ in a t becomes $\chi^2_{(2)}/2$ under the square root. Students constantly forget the division.
- **The t has a square root over the whole denominator**; the F does not.

THE RECIPE

How to answer "construct a random variable distributed as [X]" in four moves.

1. **Count what you need.** A $t_{(p)}$ needs one standard normal and one independent $\chi^2_{(p)}$. An $F_{(r_1,r_2)}$ needs two independent chi-squares with those degrees of freedom.
2. **Standardize every normal you are given:** $Z_j = (X_j - \mu_j)/\sigma_j$. You now have a pile of independent $N(0,1)$s.
3. **Assemble the chi-squares** by squaring and summing as many Z_j as you need degrees of freedom.
4. **Assemble the final ratio**, dividing each chi-squared by its own degrees of freedom, and **state explicitly which variables are independent of which**.

Critical constraint: a variable used in the numerator may **not** reappear in the denominator, or the independence requirement fails and the construction is wrong. Dr. Flores says this in the PS7 answer key in so many words.

WORKED EXAMPLE

Sample Final #5, both parts. $X \sim N(\mu_x, \sigma_x^2)$, $Y \sim N(\mu_y, \sigma_y^2)$, $Z \sim N(\mu_z, \sigma_z^2)$, *mutually independent.*

(a) What is the distribution of $\dfrac{(X-\mu_x)^2\sigma_z^2}{(Z-\mu_z)^2\sigma_x^2}$?

Do not be intimidated by the sigmas floating around. Reorganize so each squared term sits over its own variance:

$$\frac{(X-\mu_x)^2\sigma_z^2}{(Z-\mu_z)^2\sigma_x^2} = \frac{\dfrac{(X-\mu_x)^2}{\sigma_x^2}}{\dfrac{(Z-\mu_z)^2}{\sigma_z^2}} = \frac{\left(\dfrac{X-\mu_x}{\sigma_x}\right)^2}{\left(\dfrac{Z-\mu_z}{\sigma_z}\right)^2} = \frac{\chi^2_{(1)}/1}{\chi^2_{(1)}/1}$$

Numerator and denominator are each a squared standard normal, so each is $\chi^2_{(1)}$, and dividing by one degree of freedom changes nothing. They are independent because X and Z are.

Therefore

$$\boxed{\sim F_{(1,1)}}$$

(b) Using some function of X, Y and Z, derive a random variable with a $t_{(2)}$ distribution. A $t_{(2)}$ needs a standard normal on top and an independent $\chi^2_{(2)}$ on the bottom, divided by 2.
Bottom. Build the $\chi^2_{(2)}$ from the two variables you will *not* use on top:

$$W = \left(\frac{Y-\mu_y}{\sigma_y}\right)^2 + \left(\frac{Z-\mu_z}{\sigma_z}\right)^2 \sim \chi^2_{(2)}$$

because it is the sum of two independent squared standard normals.
Top. Use the remaining variable: $\dfrac{X-\mu_x}{\sigma_x} \sim N(0,1)$.
Assemble. Since X is independent of both Y and Z, it is independent of W:

$$\boxed{\frac{\dfrac{X-\mu_x}{\sigma_x}}{\sqrt{W/2}} \sim t_{(2)}} \qquad \text{where} \qquad W = \left(\frac{Y-\mu_y}{\sigma_y}\right)^2 + \left(\frac{Z-\mu_z}{\sigma_z}\right)^2$$

Notice the allocation logic: you had three variables and needed one normal plus two degrees of freedom. Spend one variable on top and the other two on the bottom. Never reuse.

WORKED EXAMPLE

PS7 #5, the same machine with a twist. $X_i \sim N(i, i^2)$ *independent for* $i = 1, 2, 3$.
Here the mean and the variance both depend on the index, so $X_i \sim N(i, i^2)$ standardizes to $(X_i - i)/i \sim N(0,1)$.
(a) $\displaystyle\sum_{i=1}^{3}\left(\frac{X_i - i}{i}\right)^2 \sim \chi^2_{(3)}$, three independent squared standard normals.
(b) Build a $t_{(2)}$. Use X_1 on top: $\dfrac{X_1 - 1}{1} = X_1 - 1 \sim N(0,1)$. Use X_2 and X_3 on the bottom:

$$\frac{X_1 - 1}{\sqrt{\dfrac{\left(\frac{X_2-2}{2}\right)^2 + \left(\frac{X_3-3}{3}\right)^2}{2}}} \sim t_{(2)}$$

The sampling-distribution versions, which are the same machine

MEMORIZE

The chi-squared can also arrive from a *sample* rather than from individual normals. Both routes appear.

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}, \qquad \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\sigma^2} \sim \chi^2_{(n)}, \qquad \frac{n(\overline{X}-\mu)^2}{\sigma^2} \sim \chi^2_{(1)}$$

Notice the degrees of freedom. Subtracting the *known* μ gives n. Subtracting the *estimated* $\overline{X}$ costs you one and gives $n-1$. Estimating a parameter from the data always costs a degree of freedom, and this is where the $n-1$ in S^2 comes from geometrically.

WORKED EXAMPLE

The t statistic you will use forever. Starting from $\overline{X} \sim N(\mu, \sigma^2/n)$:

$$\frac{\sqrt{n}(\overline{X}-\mu)}{\sigma} \sim N(0,1)$$

but σ is unknown, so replace it with S. What happens? Assemble the t:

$$\frac{\sqrt{n}(\overline{X}-\mu)}{S} = \frac{\frac{\sqrt{n}(\overline{X}-\mu)}{\sigma}}{\sqrt{\frac{(n-1)S^2/\sigma^2}{n-1}}} = \frac{N(0,1)}{\sqrt{\chi^2_{(n-1)}/(n-1)}} \sim t_{(n-1)}$$

Independence is what makes this legal, and it comes from Theorem 5.3.1(b): $\overline{X}$ and S^2 are independent under normality. Without that, the construction collapses. **Cite it.**

WORKED EXAMPLE

PS7 #4, the two-sample F. *Independent samples from $N(\mu_x, \sigma_x^2)$ and $N(\mu_y, \sigma_y^2)$. Show* $\frac{s_x^2/\sigma_x^2}{s_y^2/\sigma_y^2} \sim F_{(n-1,m-1)}$.
Each sample independently satisfies Theorem 5.3.1(c):

$$\frac{(n-1)s_x^2}{\sigma_x^2} \sim \chi^2_{(n-1)}, \qquad \frac{(m-1)s_y^2}{\sigma_y^2} \sim \chi^2_{(m-1)}$$

and these are independent because the two samples are. Now divide each by its own degrees of freedom:

$$\frac{s_x^2/\sigma_x^2}{s_y^2/\sigma_y^2} = \frac{\frac{(n-1)s_x^2}{\sigma_x^2}\cdot\frac{1}{n-1}}{\frac{(m-1)s_y^2}{\sigma_y^2}\cdot\frac{1}{m-1}} = \frac{\chi^2_{(n-1)}/(n-1)}{\chi^2_{(m-1)}/(m-1)} \sim F_{(n-1,m-1)}$$

The $(n-1)$ and $\frac{1}{n-1}$ cancel, which is exactly why the ratio of the two s^2/σ^2 terms is already in F form.

MEMORIZE

Two final relations (Theorem 5.3.8), which make good one-line exam parts:

$$X \sim F_{(p,q)} \implies \frac{1}{X} \sim F_{(q,p)} \qquad X \sim t_{(q)} \implies X^2 \sim F_{(1,q)}$$

The second explains the $F_{(1,1)}$ in Sample Final #5(a): a squared $t_{(1)}$. It also gives, from the t statistic above,

$$\frac{n(\overline{X}-\mu)^2}{S^2} \sim F_{(1,n-1)}$$

And memorize the moments: $E(\chi^2_{(r)}) = r$ and $\text{Var}(\chi^2_{(r)}) = 2r$. The variance is used constantly to back out $\text{Var}(S^2)$.

TRAP

Three ways to lose this question.

1. **Reusing a variable** on both sides of the ratio. Independence fails, the construction is invalid, and Dr. Flores calls it out explicitly in the key.
2. **Forgetting to divide the chi-squared by its degrees of freedom.** $\frac{N(0,1)}{\sqrt{\chi^2_{(2)}}}$ is not a t; $\frac{N(0,1)}{\sqrt{\chi^2_{(2)}/2}}$ is.
3. **Dividing by the variance instead of the standard deviation** when standardizing. $(X-\mu)/\sigma^2$ is not standard normal.

BLOCK 8 The score identity and the information matrix identity

30 minutes

SYMBOL DECODER

The score, one more time, because it is the object everything hangs on.

$$S(\theta, X) = \frac{\partial \ln f(X \mid \theta)}{\partial \theta}$$

It is the derivative of the log density for a **single observation**. No sum. If you sum it over the sample you get the derivative of the log-likelihood, which is a different object used for a different purpose (finding the MLE).

Why it is called the score: it measures how sensitive the log-likelihood of one data point is to moving the parameter. Where the likelihood peaks, the score is zero.

MEMORIZE

IDENTITY 1: the Score Identity.

$$E\big[S(\theta_0, X)\big] = 0$$

IDENTITY 2: the Information Matrix Identity.

$$-E\left[\frac{\partial S(\theta_0, X)}{\partial \theta}\right] = E\big[S^2(\theta_0, X)\big]$$

which written out in full is

$$-E\left[\frac{\partial^2 \ln f(X \mid \theta_0)}{\partial \theta^2}\right] = E\left[\left(\frac{\partial \ln f(X \mid \theta_0)}{\partial \theta}\right)^2\right]$$

Both hold only when everything is evaluated at θ_0, the true value. That caveat is the entire content of the exam questions about them.

WHY THIS IS TRUE

Where the score identity comes from, and why the "at θ_0" matters so much.
The proof sketch: for any θ, $f(x \mid \theta)$ is a density, so $\int f(x \mid \theta)\, dx = 1$. Differentiate both sides with respect to θ. The right side is a constant, so it gives zero. The left side, after swapping derivative and integral and multiplying and dividing by f, becomes $\int \frac{\partial \ln f(x|\theta)}{\partial \theta} f(x \mid \theta)\, dx$. That is an expectation, *but taken with respect to the density* $f(\cdot \mid \theta)$. It equals a genuine expectation over the real data only when θ happens to be the true θ_0, because the real data are generated by $f(\cdot \mid \theta_0)$.
The intuition: the score identity is a first-order condition. It can be shown that θ_0 maximizes $E[\ln f(X \mid \theta)]$ over θ. At a maximum the derivative is zero, so $E[\partial \ln f / \partial \theta]$ evaluated at θ_0 is zero. Move away from θ_0 and you are no longer at the peak, so the derivative is no longer zero.
The swap of derivative and integral is the fragile step. If the support of X depends on θ_0, the limits of the integral move when θ moves, the swap is illegal, and both identities can fail. That is Block 12.

WORKED EXAMPLE

PS8 #4, parts (b) through (d). He assigned all of 4a to 4d, so this is live. $X \sim N(0, \sigma_0^2)$, $f(x \mid \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/(2\sigma^2)}$.
(b) The score. From Block 4, $\ln f = -\frac{1}{2}\ln(2\pi) - \frac{1}{2}\ln(\sigma^2) - \frac{x^2}{2\sigma^2}$. Differentiate with respect to σ^2:

$$S(X, \sigma^2) = -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4} X^2$$

(c) Verify the score identity. Evaluate at $\sigma^2 = \sigma_0^2$ and take expectations, remembering $E(X^2) = \text{Var}(X) = \sigma_0^2$ since $E(X) = 0$:

$$E\left[S(X, \sigma_0^2)\right] = -\frac{1}{2\sigma_0^2} + \frac{1}{2\sigma_0^4} E(X^2) = -\frac{1}{2\sigma_0^2} + \frac{\sigma_0^2}{2\sigma_0^4} = -\frac{1}{2\sigma_0^2} + \frac{1}{2\sigma_0^2} = 0 \checkmark$$

(d) Verify the information matrix identity. You need both sides.
Right side, using $E(X^2) = \sigma_0^2$ and, for a normal with mean zero, $E(X^4) = 3\sigma_0^4$:

$$E\left[S^2(X, \sigma_0^2)\right] = E\left[\left(-\frac{1}{2\sigma_0^2} + \frac{X^2}{2\sigma_0^4}\right)^2\right] = E\left[\frac{1}{4\sigma_0^4} - \frac{X^2}{2\sigma_0^6} + \frac{X^4}{4\sigma_0^8}\right]$$
$$= \frac{1}{4\sigma_0^4} - \frac{\sigma_0^2}{2\sigma_0^6} + \frac{3\sigma_0^4}{4\sigma_0^8} = \frac{1}{4\sigma_0^4} - \frac{2}{4\sigma_0^4} + \frac{3}{4\sigma_0^4} = \frac{2}{4\sigma_0^4} = \frac{1}{2\sigma_0^4}$$

Left side, differentiating the score again with respect to σ^2:

$$\frac{\partial S}{\partial \sigma^2} = \frac{1}{2\sigma^4} - \frac{X^2}{(\sigma^2)^3} \quad \Longrightarrow \quad -E\left[\frac{\partial S}{\partial \sigma^2}\right] = -\frac{1}{2\sigma_0^4} + \frac{\sigma_0^2}{\sigma_0^6} = -\frac{1}{2\sigma_0^4} + \frac{2}{2\sigma_0^4} = \frac{1}{2\sigma_0^4}$$

The two sides agree, so the identity holds, and along the way you have computed $I(\theta_0) = \frac{1}{2\sigma_0^4}$, which gives CRLB $= \frac{2\sigma_0^4}{n}$.

MEMORIZE

Normal moments you may need and might not be given: for $X \sim N(0, \sigma^2)$,

$$E(X) = 0, \quad E(X^2) = \sigma^2, \quad E(X^3) = 0, \quad E(X^4) = 3\sigma^4$$

All odd moments vanish by symmetry. The fourth moment $3\sigma^4$ appeared in PS8 #4d and is worth having on your cheat sheet, since verifying the information identity for a normal variance is the one place it is essential.

WORKED EXAMPLE

A cleaner instance, from Notes XII. $X \sim N(\mu_0, 1)$.
$\ln f = -\frac{1}{2}\ln(2\pi) - \frac{(x-\mu)^2}{2}$, so $S(\mu, X) = X - \mu$. Then:

- **Score identity:** $E[S(\mu_0, X)] = E(X) - \mu_0 = \mu_0 - \mu_0 = 0$ ✓. At a general μ you would get $\mu_0 - \mu \neq 0$, which is precisely the point.
- **Information identity:** $\partial S/\partial \mu = -1$ so the left side is 1. And $E[S^2(\mu_0, X)] = E[(X - \mu_0)^2] = \text{Var}(X) = 1$. Equal ✓. At a general μ, $E[(X-\mu)^2]$ is not the variance, so equality breaks.

The lesson in one line: $E[(X-\mu)^2]$ is only the variance when μ is the true mean.

BLOCK 9 The CRLB three ways, and the UMVUE decision

45 minutes

WHY THIS IS TRUE

Why do we restrict attention to unbiased estimators at all?
Because otherwise the question "what is the best estimator?" has no answer. Dr. Flores's example: for Exponential(μ_0) data, compare $\widehat{\mu}_1 = \overline{X}$, with MSE $= \mu_0^2/n$, against the deliberately silly $\widehat{\mu}_2 = 8$, with MSE $= (8 - \mu_0)^2$. If the truth happens to be $\mu_0 = 8$, the silly estimator has zero MSE and wins. You cannot beat every estimator at every parameter value, because for any value someone can propose the constant equal to it.
So we shrink the field: consider only *unbiased* estimators, which rules out constants, and then look for the one with the smallest variance. That is the UMVUE. The cost of this restriction is real; it excludes some genuinely good slightly-biased estimators like $\widetilde{S}^2$. Asymptotics (Block 11)

is partly a response to that cost.

MEMORIZE

Definition. $\widehat{\theta}^*$ is the **Uniform Minimum Variance Unbiased Estimator** of θ_0 if
(a) $E(\widehat{\theta}^*) = \theta_0$ for all $\theta_0 \in \Theta$, and
(b) $\text{Var}(\widehat{\theta}^*) \leq \text{Var}(\widehat{\theta})$ for every other unbiased $\widehat{\theta}$ and every $\theta_0 \in \Theta$.
"Uniform" means condition (b) holds at *every* parameter value, not just at some convenient one.
Cramér-Rao Inequality (i.i.d. case). For any unbiased $\widehat{\theta}$, under regularity conditions,

$$\text{Var}(\widehat{\theta}) \;\geq\; \frac{1}{nI(\theta_0)} \;=\; \text{CRLB}$$

THE RECIPE

The UMVUE decision procedure. This is a yes/no question with a two-part test.

1. **Is it unbiased?** Compute $E(\widehat{\theta})$. If $E(\widehat{\theta}) \neq \theta_0$, **stop**. A biased estimator cannot be a UMVUE, full stop, because unbiasedness is part of the definition. Write "No: it is biased, and UMVUE requires unbiasedness." That is a complete answer.
2. **Does its variance equal the CRLB?** Compute $\text{Var}(\widehat{\theta})$ and compare with $\frac{1}{nI(\theta_0)}$.
3. **If both yes: it is the UMVUE.** Say so, and cite that no unbiased estimator can have variance below the bound.

One-directional warning: unbiased plus variance equal to CRLB *implies* UMVUE. The converse is false. An estimator can be UMVUE without attaining the bound, so if the variance exceeds the CRLB you may *not* conclude it is not UMVUE. On this exam, the questions are built so the bound is attained when the answer is yes.

WORKED EXAMPLE

Sample Final #4b, 4c, 4d. Bernoulli, the canonical version. $f(x \mid p_0) = p_0^x(1-p_0)^{1-x}$, *with* $E(X) = p_0$ *and* $\text{Var}(X) = p_0(1-p_0)$.
Score. $\ln f = x \ln p + (1-x)\ln(1-p)$, so

$$S(X,p) = \frac{X}{p} - \frac{1-X}{1-p} = \frac{X(1-p) - p(1-X)}{p(1-p)} = \frac{X-p}{p(1-p)}$$

Always simplify to this last form. Dr. Flores's handwritten note says as much: it is the easier one to work with, because it is linear in X.
Information, the easy way. Since the score is linear in X:

$$I(\theta_0) = \text{Var}\left[\frac{X-p_0}{p_0(1-p_0)}\right] = \frac{\text{Var}(X)}{p_0^2(1-p_0)^2} = \frac{p_0(1-p_0)}{p_0^2(1-p_0)^2} = \frac{1}{p_0(1-p_0)}$$

(b) The CRLB.

$$\text{CRLB} = \frac{1}{nI(\theta_0)} = \frac{p_0(1-p_0)}{n}$$

(c) Is $\widehat{p}_{MLE} = \overline{X}$ a UMVUE? Two checks. Unbiased: $E(\overline{X}) = E(X) = p_0$ ✓. Variance:

$$\text{Var}(\overline{X}) = \frac{\text{Var}(X)}{n} = \frac{p_0(1-p_0)}{n} = \text{CRLB} \checkmark$$

Yes, $\overline{X}$ is the UMVUE for p_0.
(d) Is $\overline{p} = \overline{X} + 0.25$ a UMVUE? $E(\overline{p}) = p_0 + 0.25 \neq p_0$. **No: it is biased, so it cannot be a UMVUE.** One line, full credit. Do not compute anything else.

SYMBOL DECODER

The same Bernoulli information, computed all three ways, following Dr. Flores's own supplementary note. Study this to learn which route to pick.
Route A, $\text{Var}[S]$ *(he calls this the easiest here):* shown above, one line, because the score is linear in X and you can quote $\text{Var}(X)$.
Route B, $E[S^2]$*:*

$$E\left[\frac{(X-p_0)^2}{p_0^2(1-p_0)^2}\right] = \frac{E[(X-p_0)^2]}{p_0^2(1-p_0)^2} = \frac{\text{Var}(X)}{p_0^2(1-p_0)^2} = \frac{1}{p_0(1-p_0)}$$

This works because $E[(X-p_0)^2]$ *is* the variance, since p_0 is the true mean. Note this route is identical to Route A here, which is no accident: $\text{Var}[S] = E[S^2]$ whenever $E[S] = 0$, which the score identity guarantees.
Route C, $-E[\partial S/\partial\theta]$*:* requires the quotient rule on $\frac{X-p}{p-p^2}$ and several lines of algebra before everything cancels to the same $\frac{1}{p_0(1-p_0)}$. Dr. Flores's verdict on his own worksheet: this was "the messier."
Takeaway: when the score simplifies to $\frac{X-\text{mean}}{\text{something}}$, use $\text{Var}[S]$ and you are done in one line.

WORKED EXAMPLE

PS9 #2a, Poisson. He said to expect a question like this one. $f(x \mid \theta_0) = \frac{\theta_0^x e^{-\theta_0}}{x!}$, *with* $E(X) = \theta_0$ *and* $\text{Var}(X) = \theta_0$.
$\ln f = x\ln\theta - \theta - \ln(x!)$, so $S(X,\theta) = \dfrac{X}{\theta} - 1$, which is linear in X. Therefore

$$I(\theta_0) = \text{Var}\left[\frac{X}{\theta_0} - 1\right] = \frac{\text{Var}(X)}{\theta_0^2} = \frac{\theta_0}{\theta_0^2} = \frac{1}{\theta_0} \qquad \Longrightarrow \qquad \text{CRLB} = \frac{1}{n \cdot 1/\theta_0} = \frac{\theta_0}{n}$$

Is $\widehat{\theta}_{MLE} = \overline{X}$ UMVUE? $E(\overline{X}) = \theta_0$ ✓ and $\text{Var}(\overline{X}) = \text{Var}(X)/n = \theta_0/n = \text{CRLB}$ ✓. **Yes.**

WORKED EXAMPLE

PS9 #2b, exponential in mean form. $f(x \mid \theta_0) = \frac{1}{\theta_0}e^{-x/\theta_0}$, *with* $E(X) = \theta_0$ *and* $\text{Var}(X) = \theta_0^2$.

$\ln f = -\ln\theta - \frac{x}{\theta}$, so $S(X,\theta) = -\frac{1}{\theta} + \frac{X}{\theta^2}$, linear in X. Therefore

$$I(\theta_0) = \text{Var}\left[-\frac{1}{\theta_0} + \frac{X}{\theta_0^2}\right] = \frac{\text{Var}(X)}{\theta_0^4} = \frac{\theta_0^2}{\theta_0^4} = \frac{1}{\theta_0^2} \qquad \Longrightarrow \qquad \text{CRLB} = \frac{\theta_0^2}{n}$$

$\widehat{\theta}_{MLE} = \overline{X}$ is unbiased with $\text{Var}(\overline{X}) = \theta_0^2/n = \text{CRLB}$, so **yes, UMVUE.**
Pattern worth noticing across these three examples: whenever the MLE turns out to be $\overline{X}$ and the distribution is a nice exponential-family member, $\overline{X}$ is the UMVUE and its variance equals the bound exactly. If your MLE is $\overline{X}$, expect a "yes."

TRAP

The CRLB is a bound on VARIANCE, not on MSE, and only for UNBIASED estimators. A biased estimator can absolutely have variance below the CRLB. This is not a paradox and not a violation: the bound simply does not apply to it. $\widetilde{S}^2$ is a standing example. So when a question asks whether some estimator "beats the bound," first check whether the bound applies at all.

TRAP

When the regularity conditions fail, and what to do about it. This is PS9 #3, and it is exactly the kind of thing that distinguishes a 90 from a 100.
The Cramér-Rao theorem assumes you can interchange differentiation and integration. **That fails when the support of X depends on θ_0.** The classic tells: $f(x \mid \theta_0)$ defined on $(0,\theta_0)$, or Uniform$(0,\theta_0)$.
When it fails, three things break at once: the CRLB may not bind, the score identity may fail, and the information matrix identity may fail, meaning **the three formulas for $I(\theta_0)$ can give three different answers**.
PS9 #3 in full. $f(x \mid \theta_0) = \dfrac{2x}{\theta_0^2}$ on $(0,\theta_0)$, and $\widehat{\theta}_1 = \frac{3}{2}\overline{X}$.

- $E(X) = \int_0^{\theta_0} x \cdot \frac{2x}{\theta_0^2} dx = \frac{2\theta_0}{3}$, so $E(\widehat{\theta}_1) = \frac{3}{2} \cdot \frac{2\theta_0}{3} = \theta_0$. **Unbiased.**
- $\ln f = \ln(2x) - 2\ln\theta$, so $S(X,\theta) = -\dfrac{2}{\theta}$. **The score does not involve X at all**, which is your warning siren. Then $E[S^2] = \frac{4}{\theta_0^2}$, and the problem instructs you to use the original definition:

$$\text{CRLB} = \frac{1}{nE[S^2(\theta_0, X)]} = \frac{\theta_0^2}{4n}$$

- $E(X^2) = \int_0^{\theta_0} x^2 \frac{2x}{\theta_0^2} dx = \frac{\theta_0^2}{2}$, so $\text{Var}(X) = \frac{\theta_0^2}{2} - \frac{4\theta_0^2}{9} = \frac{\theta_0^2}{18}$, giving

$$\text{Var}(\widehat{\theta}_1) = \frac{9}{4} \cdot \frac{\text{Var}(X)}{n} = \frac{9}{4} \cdot \frac{\theta_0^2}{18n} = \frac{\theta_0^2}{8n}$$

And now the punchline: $\dfrac{\theta_0^2}{8n} < \dfrac{\theta_0^2}{4n}$. An unbiased estimator sits *below* the Cramér-Rao Lower Bound. Nothing is broken; the theorem's assumptions simply do not hold, because the support $(0,\theta_0)$ moves with θ_0.

If you see a support that depends on θ, say so out loud in your answer. That sentence is very likely the point being tested.

MEMORIZE

Quick reference: score, information, and CRLB for every density in the problem sets.

Density	Score $S(\theta, X)$	$I(\theta_0)$	CRLB	MLE, UMVUE?
Bernoulli $p^x(1-p)^{1-x}$	$\dfrac{X-p}{p(1-p)}$	$\dfrac{1}{p_0(1-p_0)}$	$\dfrac{p_0(1-p_0)}{n}$	$\overline{X}$, yes
Poisson $\dfrac{\theta^x e^{-\theta}}{x!}$	$\dfrac{X}{\theta}-1$	$\dfrac{1}{\theta_0}$	$\dfrac{\theta_0}{n}$	$\overline{X}$, yes
Exponential $\dfrac{1}{\theta}e^{-x/\theta}$	$-\dfrac{1}{\theta}+\dfrac{X}{\theta^2}$	$\dfrac{1}{\theta_0^2}$	$\dfrac{\theta_0^2}{n}$	$\overline{X}$, yes
Exponential $\theta e^{-\theta x}$	$\dfrac{1}{\theta}-X$	$\dfrac{1}{\theta_0^2}$	$\dfrac{\theta_0^2}{n}$	$1/\overline{X}$, biased
$\theta x^{\theta-1}$ on $(0,1)$	$\dfrac{1}{\theta}+\ln X$	$\dfrac{1}{\theta_0^2}$	$\dfrac{\theta_0^2}{n}$	$\dfrac{-n}{\sum \ln X_i}$
$N(0,\sigma^2)$	$-\dfrac{1}{2\sigma^2}+\dfrac{X^2}{2\sigma^4}$	$\dfrac{1}{2\sigma_0^4}$	$\dfrac{2\sigma_0^4}{n}$	$\frac{1}{n}\sum X_i^2$, yes
$N(\mu,\sigma^2)$, σ^2 known	$\dfrac{X-\mu}{\sigma^2}$	$\dfrac{1}{\sigma^2}$	$\dfrac{\sigma^2}{n}$	$\overline{X}$, yes
$\dfrac{2x}{\theta^2}$ on $(0,\theta)$	$-\dfrac{2}{\theta}$	*fails*	$\theta_0^2/4n$, *not valid*	regularity fails

Notice how often $I(\theta_0)$ is just $1/\theta_0^2$ or $1/\mathrm{Var}(X)$. That is not coincidence; for these families the information is the reciprocal of the variance of the natural estimator.

BLOCK 10 Comparing estimators by mean squared error

30 minutes

MEMORIZE

$$\mathrm{MSE}(\widehat{\theta}) = E\left[(\widehat{\theta}-\theta_0)^2\right] = \mathrm{Var}(\widehat{\theta}) + \underbrace{\left[E(\widehat{\theta})-\theta_0\right]^2}_{\mathrm{Bias}^2(\widehat{\theta})}$$

Precision plus accuracy. Var measures how tightly the estimator's distribution is concentrated; Bias^2 measures how far off-center it sits. MSE penalizes both.
Immediate corollary: if $\widehat{\theta}$ is unbiased, $\mathrm{MSE}(\widehat{\theta}) = \mathrm{Var}(\widehat{\theta})$**.** Half of all MSE questions are secretly variance questions.

WHY THIS IS TRUE

Where the decomposition comes from, in case he asks you to derive it. It is one application of the variance flip.

Recall $E(Y^2) = \text{Var}(Y) + [E(Y)]^2$. Set $Y = \widehat{\theta} - \theta_0$:

$$E\left[(\widehat{\theta} - \theta_0)^2\right] = \text{Var}(\widehat{\theta} - \theta_0) + \left[E(\widehat{\theta} - \theta_0)\right]^2$$

Now θ_0 is a constant, so subtracting it does not change the variance: $\text{Var}(\widehat{\theta} - \theta_0) = \text{Var}(\widehat{\theta})$. And $E(\widehat{\theta} - \theta_0) = E(\widehat{\theta}) - \theta_0 = \text{Bias}(\widehat{\theta})$. Substituting gives the result. Two lines.

THE RECIPE

How to answer "which estimator has lower MSE?"

1. For each estimator: compute $E(\widehat{\theta})$, hence the bias.
2. For each estimator: compute $\text{Var}(\widehat{\theta})$, using $\text{Var}(a + bY) = b^2\text{Var}(Y)$ and $\text{Var}(\overline{X}) = \sigma^2/n$.
3. Add: $\text{MSE} = \text{Var} + \text{Bias}^2$.
4. Compare. **If the comparison depends on n or on θ_0, say so explicitly and give the condition.** That is often the actual point of the question.

WORKED EXAMPLE

Sample Final #2. *i.i.d. sample, unknown mean μ_0, known variance $\sigma^2 = 1$. Compare $\widehat{\mu} = \overline{X} + 1$ against $\widetilde{\mu} = (X_1 + X_2)/2$.*
First estimator. $E(\widehat{\mu}) = \mu_0 + 1$, so Bias $= 1$. And $\text{Var}(\widehat{\mu}) = \text{Var}(\overline{X}) = \frac{\sigma^2}{n} = \frac{1}{n}$, since adding a constant does not change variance. Therefore

$$\text{MSE}(\widehat{\mu}) = \frac{1}{n} + 1^2 = \frac{1}{n} + 1$$

Second estimator. $E(\widetilde{\mu}) = \frac{\mu_0 + \mu_0}{2} = \mu_0$, so it is unbiased and MSE $=$ Var. And

$$\text{Var}(\widetilde{\mu}) = \frac{1}{4}\left[\text{Var}(X_1) + \text{Var}(X_2)\right] = \frac{1}{4}(1 + 1) = \frac{1}{2}$$

so $\text{MSE}(\widetilde{\mu}) = \frac{1}{2}$.
Compare. $\text{MSE}(\widehat{\mu}) = \frac{1}{n} + 1 > 1 > \frac{1}{2} = \text{MSE}(\widetilde{\mu})$ for every n. **$\widetilde{\mu}$ has lower MSE.**
The moral, which is the pedagogical point of the question: $\widetilde{\mu}$ throws away all but two observations, which is wasteful, and yet it still wins, because $\widehat{\mu}$ carries a bias of 1 that never shrinks no matter how much data you collect. **A bias that does not vanish with n is fatal.** Variance can be beaten down by sample size; a fixed bias cannot.

WORKED EXAMPLE

PS9 #1, the harder version. He warned this variance is not straightforward, and said an exam version would be easier. *$(X_1, \ldots, X_n) \overset{iid}{\sim} N(\mu_0, \sigma_0^2)$ with μ_0 known. Compare $v^2 = \frac{1}{n}\sum X_i^2$ against $s^2 = \frac{1}{n-1}\sum(X_i - \overline{X})^2$ as estimators of σ_0^2.* (Here $\mu_0 = 0$, as in the problem set.)
Unbiasedness. $E(v^2) = \frac{1}{n}\sum E(X_i^2) = \frac{1}{n} \cdot n\sigma_0^2 = \sigma_0^2$, using $E(X_i^2) = \text{Var}(X_i) = \sigma_0^2$ because the mean is zero. And $E(s^2) = \sigma_0^2$ by Theorem 5.2.6(c). **Both unbiased, so both MSEs are just**

variances.
Variance of s^2. From PS7 #3, $\text{Var}(s^2) = \dfrac{2\sigma_0^4}{n-1}$.
Variance of v^2. Since $X_i/\sigma_0 \sim N(0,1)$ and the X_i are independent,

$$\frac{nv^2}{\sigma_0^2} = \sum_{i=1}^{n}\left(\frac{X_i}{\sigma_0}\right)^2 \sim \chi^2_{(n)}$$

Apply $\text{Var}(\chi^2_{(n)}) = 2n$ and pull constants out:

$$2n = \text{Var}\left(\frac{nv^2}{\sigma_0^2}\right) = \frac{n^2}{\sigma_0^4}\text{Var}(v^2) \quad \Longrightarrow \quad \text{Var}(v^2) = \frac{2\sigma_0^4}{n}$$

Compare. $\dfrac{2\sigma_0^4}{n-1} > \dfrac{2\sigma_0^4}{n}$, since $n-1 < n$. v^2 **has lower MSE.**
Why v^2 **wins:** it uses the known value of the mean instead of estimating it with $\overline{X}$. Estimating the mean costs a degree of freedom, which shows up as $n-1$ in the denominator and inflates the variance. **Knowing more gives you a better estimator.** The technique of the middle step, recognizing a chi-squared and back-solving for the variance, is worth memorizing as a standalone move.

MEMORIZE

The two variance-extraction moves used above, because they show up repeatedly:

$$\text{Var}(a+bY) = b^2\text{Var}(Y) \qquad \text{If } cY \sim \chi^2_{(r)} \text{ then } \text{Var}(Y) = \frac{2r}{c^2}$$

The second is how PS7 #3, PS9 #1, and PS10 #1 all get their variances. Recognize a chi-squared, apply Var = 2 df, and divide out the constant squared.

WORKED EXAMPLE

The bias-variance trade-off, S^2 **against** $\widetilde{S}^2$**.** A natural exam question, and the reason the course cares about MSE at all.
S^2 is unbiased with $\text{Var}(S^2) = \frac{2\sigma^4}{n-1}$, so $\text{MSE}(S^2) = \frac{2\sigma^4}{n-1}$.
$\widetilde{S}^2 = \frac{n-1}{n}S^2$ is biased: $E(\widetilde{S}^2) = \frac{n-1}{n}\sigma^2$, so Bias $= -\frac{\sigma^2}{n}$. Its variance is

$$\text{Var}(\widetilde{S}^2) = \left(\frac{n-1}{n}\right)^2 \frac{2\sigma^4}{n-1} = \frac{2(n-1)\sigma^4}{n^2}$$

so

$$\text{MSE}(\widetilde{S}^2) = \frac{2(n-1)\sigma^4}{n^2} + \frac{\sigma^4}{n^2} = \frac{(2n-1)\sigma^4}{n^2}$$

Compare with $\text{MSE}(S^2) = \frac{2\sigma^4}{n-1} = \frac{2n\sigma^4}{n(n-1)}$. Since $\frac{2n-1}{n^2} < \frac{2}{n-1}$ for all $n \geq 2$, **the biased estimator** $\widetilde{S}^2$ **actually has the lower MSE.**
This is the whole argument for why restricting to unbiased estimators has a cost. $\widetilde{S}^2$ is excluded from UMVUE consideration by definition, yet it beats the UMVUE on the criterion

we actually care about.

BLOCK 11 Convergence, WLLN, CLT, Slutsky, delta method

40 minutes

WHY THIS IS TRUE

Why this chapter exists. Two problems with everything before it.
First, unbiased estimators are hard to find and fragile. $E(S^2) = \sigma^2$ but $E(S) \neq \sigma$, so the moment you want a function of a parameter, unbiasedness evaporates. Second, we usually cannot derive the exact distribution of an estimator, which we need in order to say anything about uncertainty.
Asymptotics solves both by asking what happens as $n \to \infty$. Consistency replaces unbiasedness and, crucially, *survives continuous transformation*. And the CLT gives an approximate normal distribution for essentially any estimator built out of sample averages, which is essentially all of them.

The three convergence concepts, consolidated

Concept	Definition	What it is for
Mean square, $\xrightarrow{q.m.}$	$\lim_{n\to\infty} E[(X_n - c)^2] = 0$	The *tool*. Splits into $\lim \text{Var} = 0$ and $\lim \text{Bias} = 0$. You use it to prove the next one.
Probability, $\xrightarrow{p}$	$\lim_{n\to\infty} P(\lvert X_n - c\rvert > \epsilon) = 0$ for all $\epsilon > 0$	The *property*. This is what consistency means. Most used concept in econometrics.
Distribution, $\xrightarrow{d}$	$\lim_{n\to\infty} F_{X_n}(r) = F_Y(r)$	The *approximation*. Gives you a usable distribution for inference.

$$\boxed{\xrightarrow{q.m.} \implies \xrightarrow{p} \implies \xrightarrow{d}}$$ and none of the reverses hold.

MEMORIZE

Weak Law of Large Numbers. For an i.i.d. sample with $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$:

$$\overline{X} \xrightarrow{p} \mu$$

Proof in three lines, via Chebyshev's inequality:

$$P(|\overline{X} - \mu| > \epsilon) \leq \frac{\text{Var}(\overline{X})}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \xrightarrow[n\to\infty]{} 0$$

so $\lim_{n\to\infty} P(|\overline{X} - \mu| > \epsilon) = 0$, which is the definition. **Know this proof.** It is short, it has appeared, and it shows exactly why consistency is really a statement about the variance shrinking.

MEMORIZE

Central Limit Theorem (Lindeberg-Lévy). For an i.i.d. sample with mean μ and variance σ^2:

$$\sqrt{n}\,(\overline{X}-\mu) \xrightarrow{d} N(0,\sigma^2) \qquad \text{equivalently} \qquad \frac{\sqrt{n}(\overline{X}-\mu)}{\sigma} \xrightarrow{d} N(0,1)$$

which gives the **asymptotic distribution**

$$\overline{X} \overset{a}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$$

Read the second display carefully. Before, under normality, we had $\overline{X} \sim N(\mu, \sigma^2/n)$ *exactly*. Now, from *any* distribution whatsoever, we get the same thing *approximately*. That is the miracle. The price is that it is an approximation valid for large n.
And note what the CLT is about: it is a statement about $\overline{X}$, not about the individual X_i. The data do not become normal. Their average does.

SYMBOL DECODER

Why the $\sqrt{n}$? Because without it the statement is boring and with a larger power it blows up. $\overline{X}-\mu$ by itself converges to zero, so its limiting distribution is a point mass at zero, which tells you nothing. Multiplying by $\sqrt{n}$ exactly offsets the shrinking: $\text{Var}[\sqrt{n}(\overline{X}-\mu)] = n \cdot \frac{\sigma^2}{n} = \sigma^2$, a constant. **The $\sqrt{n}$ is the unique inflation rate that stabilizes the variance**, which is why it appears in every limiting-distribution statement in this course.

MEMORIZE

Useful asymptotic results. Let a, b be constants.
(a) Arithmetic of probability limits. If $X_n \xrightarrow{p} a$ and $Y_n \xrightarrow{p} b$:

$$X_n + Y_n \xrightarrow{p} a+b, \qquad X_n Y_n \xrightarrow{p} ab, \qquad \frac{X_n}{Y_n} \xrightarrow{p} \frac{a}{b} \quad (b \neq 0)$$

(b) Slutsky's Theorem. If $X_n \xrightarrow{d} U$ (a random variable) and $Y_n \xrightarrow{p} a$ (a constant):

$$X_n Y_n \xrightarrow{d} aU, \qquad X_n + Y_n \xrightarrow{d} U + a, \qquad \frac{X_n}{Y_n} \xrightarrow{d} \frac{U}{a} \quad (a \neq 0)$$

(c) Continuous mapping. If $\widehat{\theta} \xrightarrow{p} \theta_0$ and g is continuous, then $g(\widehat{\theta}) \xrightarrow{p} g(\theta_0)$.
(d) Delta method. If $\sqrt{n}(\widehat{\theta}-\theta_0) \xrightarrow{d} N(0,\Lambda)$ and $g'(\theta_0) \neq 0$:

$$\sqrt{n}\left[g(\widehat{\theta}) - g(\theta_0)\right] \xrightarrow{d} N\big(0, [g'(\theta_0)]^2\Lambda\big)$$

SYMBOL DECODER

Slutsky versus the delta method: which one does this question want?

Slutsky	You are *combining* two sequences, one converging in distribution and one converging in probability to a constant. Signature: the question hands you two separate objects, Y and Z, and asks about YZ or $Y+Z$ or Y/Z.
Delta method	You are *transforming* one estimator through a function. Signature: the question says "estimate $g(\theta_0)$" or "find the asymptotic distribution of $\sqrt{\widehat{\theta}}$ / $e^{\widehat{\theta}}$ / $1/\widehat{\theta}$."

One-word test: combining two things means Slutsky, transforming one thing means delta.

WORKED EXAMPLE

Sample Final #6, a pure Slutsky question. $Y \xrightarrow{d} N(0, \sigma^2\Omega)$ *with* $\sigma^2 > 0$ *and* $\Omega > 0$, *and* $Z \xrightarrow{p} \Omega^{-1}$. *What is the limiting distribution of* YZ?

Two separate objects being multiplied, one converging in distribution and one in probability to a constant. That is Slutsky part (i): $X_n Y_n \xrightarrow{d} aU$.

Here $a = \Omega^{-1}$ and $U \sim N(0, \sigma^2\Omega)$, so

$$YZ \xrightarrow{d} \Omega^{-1} \cdot N(0, \sigma^2\Omega)$$

Now simplify. Multiplying a normal by a constant multiplies its variance by the constant *squared*:

$$\text{Var}\left[\Omega^{-1}U\right] = \Omega^{-2} \cdot \sigma^2\Omega = \sigma^2\Omega^{-1}$$

Therefore

$$\boxed{YZ \xrightarrow{d} N\left(0,\ \sigma^2\Omega^{-1}\right)}$$

The step people drop is squaring the constant. Ω^{-1} outside a normal becomes Ω^{-2} inside the variance, and only then does one power of Ω cancel.

WORKED EXAMPLE

A Slutsky question in its natural habitat. *Suppose* $\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, V)$ *and* $\widehat{\sigma}^2 \xrightarrow{p} \sigma^2 > 0$. *What is the limiting distribution of* $\dfrac{\sqrt{n}(\widehat{\theta} - \theta_0)}{\widehat{\sigma}}$?

By continuous mapping, $\widehat{\sigma} \xrightarrow{p} \sigma$. Then Slutsky part (iii):

$$\frac{\sqrt{n}(\widehat{\theta} - \theta_0)}{\widehat{\sigma}} \xrightarrow{d} \frac{N(0, V)}{\sigma} = N\left(0, \frac{V}{\sigma^2}\right)$$

This is the theoretical justification for every t-statistic you will ever compute in econometrics: replacing an unknown standard deviation by a consistent estimate of it does not change the limiting distribution. That is why Slutsky matters far beyond this exam.

MEMORIZE

Asymptotic efficiency, and the theorem that ends the course.
A sequence $\{\widehat{\theta}_n\}$ is **asymptotically efficient** for θ_0 if

$$\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N\Big(0,\ [I(\theta_0)]^{-1}\Big) \qquad \text{equivalently} \qquad \widehat{\theta} \overset{a}{\sim} N\left(\theta_0,\ \frac{1}{nI(\theta_0)}\right) = N(\theta_0,\ \text{CRLB})$$

So asymptotic efficiency means: asymptotically unbiased, and asymptotic variance equal to the CRLB. It is the large-sample analogue of UMVUE.
Theorem (Asymptotic Properties of the MLE). Under regularity conditions, $\widehat{\theta}_{MLE}$ is

(a) **consistent**: $\widehat{\theta}_{MLE} \xrightarrow{p} \theta_0$

(b) **asymptotically normal**: $\sqrt{n}(\widehat{\theta}_{MLE} - \theta_0) \xrightarrow{d} N(0, [I(\theta_0)]^{-1})$

(c) **asymptotically efficient**: $\widehat{\theta}_{MLE} \overset{a}{\sim} N(\theta_0, [nI(\theta_0)]^{-1})$

This single theorem is why the pipeline in Block 5 works. And it is why PS10 #4a can ask for "an estimator that is consistent, asymptotically normal and asymptotically efficient" and the answer is simply "the MLE, by this theorem." If you see that phrase, name the theorem and write down the MLE.

TRAP

Asymptotically unbiased is not the same as unbiased. $\widehat{\theta}_{MLE}$ is frequently biased in finite samples. Dr. Flores says most MLEs are. Its good behavior is entirely a large-sample story. So an exam answer that says "the MLE is unbiased" is wrong; "the MLE is consistent and asymptotically unbiased" is right.

BLOCK 12 Every trap, and the strict-grading checklist 25 minutes

The nine traps, ranked by how likely they are to cost you a point

T1. Putting a summation inside the score. $S(\theta, X)$ is one observation. The n enters only later, in CRLB $= 1/(nI(\theta_0))$. If your $I(\theta_0)$ has an n in it, you made this error.

T2. Differentiating with respect to σ when the parameter is σ^2. Rename $v = \sigma^2$ and work in v.

T3. Confusing S^2 (divide by $n-1$, unbiased) with $\widetilde{S}^2$ (divide by n, MOM and MLE, biased). Read which one the problem defines and use that one throughout.

T4. Mixing centered and uncentered moments in MOM. Pair $E(X)$ with $\overline{X}$, and $\mathrm{Var}(X)$ with $\widetilde{S}^2$. Never pair $\mathrm{Var}(X)$ with $\frac{1}{n}\sum X_i^2$.

T5. Applying Theorem 5.3.1 without normality. $E(\overline{X}) = \mu$, $\mathrm{Var}(\overline{X}) = \sigma^2/n$, and $E(S^2) = \sigma^2$ hold always. "$\overline{X} \sim N$" and "$(n-1)S^2/\sigma^2 \sim \chi^2$" and "$\overline{X} \perp S^2$" need normality. State the assumption when you use it.

T6. Reusing a random variable in numerator and denominator of a t or F construction, destroying independence.

T7. Forgetting to divide a chi-squared by its own degrees of freedom inside a t or F.

T8. Forgetting to square $g'(\theta_0)$ in the delta method, or forgetting to square the constant in Slutsky.

T9. Applying the CRLB when the support depends on θ_0. If you see $(0, \theta_0)$ or $\mathrm{Uniform}(0, \theta_0)$, the regularity conditions fail. Say so.

TRAP

The support trap, expanded, because it appears in three different disguises.
Disguise 1, MLE by calculus. $\mathrm{Uniform}(0, \theta_0)$: the first-order condition has no solution and $\widehat{\theta}_{MLE} = \max\{X_i\}$ instead. (PS8 #3b)
Disguise 2, CRLB violated. $f = 2x/\theta_0^2$ on $(0, \theta_0)$: an unbiased estimator has variance below the "bound." (PS9 #3)
Disguise 3, the three information formulas disagree. When the interchange of derivative and integral fails, $E[S^2]$, $\mathrm{Var}[S]$, and $-E[\partial S/\partial\theta]$ can give three different numbers. Use the definitional one, $E[S^2]$, and note the caveat.
Your detector: does the range of x in the density contain θ? If yes, stop and flag it before doing anything else.

TRAP

The parameterization trap. The exponential distribution appears in two forms in this course and they give reciprocal answers.

Form	$E(X)$	$\mathrm{Var}(X)$	MLE
$\frac{1}{\theta}e^{-x/\theta}$ (mean / scale)	θ	θ^2	$\overline{X}$
$\theta e^{-\theta x}$ (rate)	$1/\theta$	$1/\theta^2$	$1/\overline{X}$

Always read the density before writing anything. The same is true of the Geometric, where $E(X) = 1/p_0$ and so $\overline{X}$ is *not* unbiased for p_0, which is exactly what Sample Final #3a tests.

The strict-grading checklist

THE RECIPE

Before you hand in any answer, verify these.

1. **Did I say "by independence" where I used it?** In every variance-of-a-sum step.
2. **Did I say "since the sample is identically distributed" where I replaced $E(X_i)$ by μ?**
3. **Did I name the theorem?** "By Theorem 5.3.1(c)", "by the invariance property of the MLE", "by Slutsky's theorem", "by the delta method", "by the asymptotic properties of the MLE." Naming costs four words and reads as mastery.
4. **Did I evaluate at θ_0?** Every identity and every information calculation.
5. **Did I give the form that was asked for?** Limiting distribution has $\sqrt{n}$ and is centered at zero. Asymptotic distribution divides by n and centers at θ_0. When unsure, write both.
6. **Did I put a hat on my estimator and a subscript naming the method?** $\widehat{\theta}_{MLE}$, $\widehat{\theta}_{MOM}$.
7. **Did I answer the question that was asked?** If it says "is it a UMVUE?", the answer begins with "Yes" or "No" and then justifies.

EXAM INTEL FROM DR. FLORES

Time strategy for the exam. He has scheduled 10:10 to 2:00 but says he is aiming for something closer to two hours. That means **time pressure is not your binding constraint**, which changes the optimal strategy: there is no reason to leave anything blank, and every reason to write out full justifications.
Suggested passes:

- **First pass, 40%:** everything you recognize instantly. MOM, MLE, the $\chi^2/t/F$ construction, the proofs from Block 2.
- **Second pass, 40%:** the multi-part estimator question, which will be worth several points. Score, information, CRLB, UMVUE, limiting distribution, delta method, all hanging off one density.
- **Third pass, 20%:** whatever remains, plus writing labeled setups for anything unfinished. **An unfinished question with a correct labeled setup is worth far more than a blank**, because his stated grading philosophy is that using the right formulas and processes matters more than algebraic correctness.

BLOCK 13 The real Sample Final, fully solved

50 minutes

This is Dr. Flores's "Sample Questions from Old Final Exams" handout, worked completely. Cover each solution, attempt the question, then uncover. Where a solution appeared earlier in this guide it is given in condensed form with a pointer.

EXAM INTEL FROM DR. FLORES

His instruction at the top of the handout: *"This is a closed-book exam. Please show your work! Try to show as much work and logic as possible."* Combined with his note that process beats algebra, the strategy is settled. **Write more, not less.**

Question 1

Q1 (1 point)

Let $X_1, \ldots, X_n$ be an i.i.d. sample with mean μ and variance σ^2. Prove $E(\overline{X}) = \mu$ and $\mathrm{Var}(\overline{X}) = \sigma^2/n$. Be clear about each step.

SOLUTION

Worked in full in Block 2. The two chains, with their justifications:

$$E(\overline{X}) = E\left[\tfrac{1}{n}\textstyle\sum X_i\right] \overset{\text{linearity}}{=} \tfrac{1}{n}\textstyle\sum E(X_i) \overset{\text{i.d.}}{=} \tfrac{1}{n}\cdot n\mu = \mu$$

$$\mathrm{Var}(\overline{X}) = \mathrm{Var}\left(\tfrac{1}{n}\textstyle\sum X_i\right) \overset{\mathrm{Var}(aY)=a^2\mathrm{Var}Y}{=} \tfrac{1}{n^2}\mathrm{Var}\left(\textstyle\sum X_i\right) \overset{\textbf{independence}}{=} \tfrac{1}{n^2}\textstyle\sum \mathrm{Var}(X_i) \overset{\text{i.d.}}{=} \tfrac{1}{n^2}n\sigma^2 = \tfrac{\sigma^2}{n}$$

The graded content is the annotations. Name linearity, name the a^2 rule, and above all name independence at the third step of the variance chain.

Question 2

Q2 (1 point)

i.i.d. sample, unknown mean μ_0, known variance $\sigma^2 = 1$. Two estimators: $\widehat{\mu} = \overline{X} + 1$ and $\widetilde{\mu} = (X_1 + X_2)/2$. Which has lower MSE?

SOLUTION

$$\mathrm{MSE}(\widehat{\mu}) = \mathrm{Var}(\overline{X}) + \mathrm{Bias}^2 = \frac{1}{n} + 1 \qquad \mathrm{MSE}(\widetilde{\mu}) = \mathrm{Var}\left(\frac{X_1 + X_2}{2}\right) + 0 = \frac{1+1}{4} = \frac{1}{2}$$

Since $\frac{1}{n} + 1 > 1 > \frac{1}{2}$ for every n, $\widetilde{\mu}$ **has the lower MSE.**
Worth saying in your answer: $\widehat{\mu}$ carries a bias of one that does not shrink with n, whereas $\widetilde{\mu}$, despite using only two observations, is unbiased. A non-vanishing bias beats a large variance.

Question 3

Q3 (2 points)

Random sample from Geometric with $f(x \mid p_0) = p_0(1-p_0)^{x-1}$, $0 < p_0 < 1$. *Given* $E(X) = 1/p_0$ *and* $\text{Var}(X) = (1-p_0)/p_0^2$. ***(a)*** *Is* $\overline{X}$ *an unbiased estimator of* p_0*?* ***(b)*** *Find the MOM estimator of* p_0*, using only the first population moment.*

SOLUTION

(a) By Theorem 5.2.6(a), $E(\overline{X}) = E(X) = \dfrac{1}{p_0}$.

The question is whether that equals p_0. It does not: $\dfrac{1}{p_0} \neq p_0$ except at the single point $p_0 = 1$, which is excluded. **No, $\overline{X}$ is biased for p_0.**

Why this question exists: it is checking whether you read the parameterization. $\overline{X}$ is unbiased for $E(X)$, always. It is unbiased for the *parameter* only when the parameter happens to be the mean. Here the mean is the reciprocal of the parameter.

(b) One parameter, so one equation. Set the population first moment equal to the sample first moment:

$$E(X) = \frac{1}{p} = \overline{X} \quad \Longrightarrow \quad \boxed{\widetilde{p}_{MOM} = \frac{1}{\overline{X}}}$$

Question 4

Q4 (7 parts, likely 3 to 4 points)

Random sample from Bernoulli, $f(x \mid p_0) = p_0^x(1-p_0)^{1-x}$ *for* $x \in \{0,1\}$, *with* $E(X) = p_0$ *and* $\text{Var}(X) = p_0(1-p_0)$.

EXAM INTEL FROM DR. FLORES

This is the single most important question in the entire handout. It is one density with seven parts hanging off it, and those seven parts cover MLE, CRLB, UMVUE twice, limiting distribution, variance estimation, invariance, and the delta method. Dr. Flores flagged four separate exam questions that look like pieces of it. **If you can do all of Q4 from a blank page, you can do most of the exam.**

(a) (Find the MLE for p_0)

SOLUTION

$$\ln f(x \mid p) = x \ln p + (1-x)\ln(1-p)$$

$$L(p) = \sum_{i=1}^{n} [X_i \ln p + (1-X_i)\ln(1-p)]$$

$$\frac{\partial L}{\partial p} = \frac{\sum X_i}{p} - \frac{n - \sum X_i}{1-p} = 0$$

Cross-multiply: $(1-p)\sum X_i = p\,(n - \sum X_i)$, so $\sum X_i - p\sum X_i = pn - p\sum X_i$, so $\sum X_i = pn$, giving

$$\boxed{\widehat{p}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} X_i = \overline{X}}$$

(b) (Find the CRLB for any unbiased estimator of p_0)

SOLUTION

Score, simplified over a common denominator:

$$S(X,p) = \frac{X}{p} - \frac{1-X}{1-p} = \frac{X(1-p) - p(1-X)}{p(1-p)} = \frac{X-p}{p(1-p)}$$

Fisher information, using $\mathrm{Var}[S]$ since the score is linear in X:

$$I(\theta_0) = \mathrm{Var}\left[\frac{X-p_0}{p_0(1-p_0)}\right] = \frac{\mathrm{Var}(X)}{p_0^2(1-p_0)^2} = \frac{p_0(1-p_0)}{p_0^2(1-p_0)^2} = \frac{1}{p_0(1-p_0)}$$

Therefore

$$\boxed{\mathrm{CRLB} = \frac{1}{nI(\theta_0)} = \frac{p_0(1-p_0)}{n}}$$

(c) (Is your estimator in (a) a UMVUE?)

SOLUTION

Two-part test.

- **Unbiased?** $E(\widehat{p}_{MLE}) = E(\overline{X}) = E(X) = p_0$. ✓
- **Variance equals the bound?** $\mathrm{Var}(\overline{X}) = \dfrac{\mathrm{Var}(X)}{n} = \dfrac{p_0(1-p_0)}{n} = \mathrm{CRLB}$. ✓

Yes. Since $\widehat{p}_{MLE}$ is unbiased and attains the Cramér-Rao Lower Bound, no unbiased estimator can have smaller variance, so $\widehat{p}_{MLE} = \overline{X}$ is the UMVUE for p_0.

(d) (Let $\overline{p} = \overline{X} + 0.25$. Is $\overline{p}$ a UMVUE?)

SOLUTION

$$E(\overline{p}) = E(\overline{X}) + 0.25 = p_0 + 0.25 \neq p_0$$

No. $\overline{p}$ is biased, and unbiasedness is part of the definition of a UMVUE, so it cannot be one. **Stop there.** Do not compute its variance. This is a one-line question and the grader is checking whether you know that unbiasedness is a prerequisite, not a bonus.

(e) (Obtain the limiting distribution of your estimator in part (a))

SOLUTION

By the theorem on asymptotic properties of the MLE, $\sqrt{n}(\widehat{p}_{MLE} - p_0) \xrightarrow{d} N(0, [I(\theta_0)]^{-1})$, and from part (b), $[I(\theta_0)]^{-1} = p_0(1-p_0)$. So

$$\boxed{\sqrt{n}\,(\widehat{p}_{MLE} - p_0) \xrightarrow{d} N(0,\ p_0(1-p_0))}$$

and therefore the asymptotic distribution is

$$\widehat{p}_{MLE} \overset{a}{\sim} N\left(p_0,\ \frac{p_0(1-p_0)}{n}\right)$$

Give both lines. The question says "limiting," so the first is what is being asked, but the second costs nothing and is what part (f) needs.
Alternative route worth mentioning: since $\widehat{p}_{MLE} = \overline{X}$, you could get the same answer straight from the CLT with $\sigma^2 = \text{Var}(X) = p_0(1-p_0)$. Both are correct.

(f) (Based on (e), how would you estimate the variance of your estimator?)

SOLUTION

From (e), $\text{AVar}(\widehat{p}_{MLE}) = \dfrac{p_0(1-p_0)}{n}$, which is a function of the unknown p_0. Replace p_0 by its MLE, justified by the invariance property:

$$\boxed{\widehat{\text{AVar}}(\widehat{p}_{MLE}) = \frac{\widehat{p}_{MLE}(1-\widehat{p}_{MLE})}{n} = \frac{\overline{X}(1-\overline{X})}{n}}$$

The sentence that earns the point: "by the invariance property of the MLE, the MLE of a function of p_0 is that function evaluated at $\widehat{p}_{MLE}$."

(g) (Suppose you want to estimate $\gamma_0 = 1/p_0$. Find the MLE and derive its asymptotic distribution)

SOLUTION

The estimator, by invariance:

$$\widehat{\gamma}_{MLE} = \frac{1}{\widehat{p}_{MLE}} = \frac{1}{\overline{X}}$$

Its distribution, by the delta method. With $g(p) = 1/p$:

$$g'(p) = -\frac{1}{p^2} \qquad \Longrightarrow \qquad [g'(p_0)]^2 = \frac{1}{p_0^4}$$

From (e), $\Lambda = p_0(1-p_0)$. Multiply:

$$[g'(p_0)]^2\Lambda = \frac{p_0(1-p_0)}{p_0^4} = \frac{1-p_0}{p_0^3}$$

Therefore

$$\boxed{\sqrt{n}\,(\widehat{\gamma}_{MLE} - \gamma_0) \xrightarrow{d} N\left(0,\ \frac{1-p_0}{p_0^3}\right)} \qquad \text{and} \qquad \widehat{\gamma}_{MLE} \overset{a}{\sim} N\left(\frac{1}{p_0},\ \frac{1-p_0}{np_0^3}\right)$$

Question 5

Q5 (2 points)

$X \sim N(\mu_x, \sigma_x^2)$, $Y \sim N(\mu_y, \sigma_y^2)$, $Z \sim N(\mu_z, \sigma_z^2)$, *mutually independent.* ***(a)*** *What is the distribution of* $\dfrac{(X-\mu_x)^2\sigma_z^2}{(Z-\mu_z)^2\sigma_x^2}$*?* ***(b)*** *Using some function of* X, Y, Z*, derive a random variable with a* $t_{(2)}$ *distribution.*

SOLUTION

Worked in full in Block 7. In brief:
(a) Regroup so each square sits over its own variance. Numerator and denominator are each $\left(\frac{\cdot-\mu}{\sigma}\right)^2 \sim \chi^2_{(1)}$, independent, each divided by one degree of freedom. Hence $\sim F_{(1,1)}$.
(b) Let $W = \left(\dfrac{Y-\mu_y}{\sigma_y}\right)^2 + \left(\dfrac{Z-\mu_z}{\sigma_z}\right)^2 \sim \chi^2_{(2)}$, a sum of two independent squared standard normals. Since X is independent of W,

$$\frac{(X-\mu_x)/\sigma_x}{\sqrt{W/2}} \sim t_{(2)}$$

State the independence explicitly, and note that X appears only in the numerator and Y, Z only in the denominator.

Question 6

Q6 (1 point)

$Y \xrightarrow{d} N(0, \sigma^2\Omega)$ *with* $\sigma^2 > 0$ *and* $\Omega > 0$*, and* $Z \xrightarrow{p} \Omega^{-1}$*. What is the limiting distribution of* YZ*?*

SOLUTION

Two objects being multiplied, one converging in distribution and one in probability to a constant, so this is Slutsky's theorem:

$$YZ \xrightarrow{d} \Omega^{-1} \cdot N(0, \sigma^2\Omega)$$

Pulling the constant inside squares it in the variance:

$$\boxed{YZ \xrightarrow{d} N\left(0,\ \sigma^2\Omega^{-1}\right)}$$

Name Slutsky's theorem in your answer.

MEMORIZE

Tally of the Sample Final by machine. Six questions, thirteen parts:

- Sampling distribution proofs: Q1
- MSE comparison: Q2
- Unbiasedness check: Q3a, Q4c, Q4d
- MOM: Q3b
- MLE: Q4a
- Score and information: Q4b
- CRLB and UMVUE: Q4b, Q4c, Q4d
- Limiting and asymptotic distributions: Q4e
- Invariance and variance estimation: Q4f, Q4g
- Delta method: Q4g
- $\chi^2/t/F$ construction: Q5a, Q5b
- Slutsky: Q6

That is the entire syllabus of the second half in thirteen parts. If you can do this handout cold, you are prepared.

BLOCK 14 Mock final: ten questions, thirteen parts

60 minutes

Built to Dr. Flores's blueprint: ten questions, thirteen scored parts, no "Hard" items, every "for sure" topic represented, and nothing from hypothesis testing or confidence intervals. Every numerical answer has been symbolically verified. Take it closed-book with your handwritten cheat sheets, give yourself sixty minutes, then grade against the solutions that follow.

Throughout, $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ denotes the sample mean and $\widetilde{S}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2$.

Question 1 (1 point)

Let $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ be two *independent* random samples from populations with means and variances (μ_x, σ_x^2) and (μ_y, σ_y^2). Prove that

$$E(\overline{X} - \overline{Y}) = \mu_x - \mu_y \qquad \text{and} \qquad \operatorname{Var}(\overline{X} - \overline{Y}) = \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}$$

Be clear about each step.

Question 2 (1 point)

Let $X_1, \ldots, X_n$ be an i.i.d. sample from a distribution with unknown mean μ_0 and known variance $\sigma^2 = 4$. Consider

$$\widehat{\mu} = \overline{X} \qquad \text{and} \qquad \widetilde{\mu} = \frac{X_1 + X_2 + X_3}{3} + 0.5$$

Which estimator has the lower mean squared error?

Question 3 (1 point)

Let $X_1, \ldots, X_n$ be a random sample from a distribution with pdf

$$f(x \mid \theta_0) = (\theta_0 + 1)x^{\theta_0}, \qquad 0 < x < 1, \quad \theta_0 > -1$$

and zero otherwise. Find the method of moments estimator of θ_0.

Question 4 (5 parts, 4 points)

Let $X_1, \ldots, X_n$ be a random sample from a distribution with pdf

$$f(x \mid \theta_0) = \frac{x}{\theta_0}\, e^{-x^2/(2\theta_0)}, \qquad x > 0, \quad \theta_0 > 0$$

and zero otherwise. You may take as given that $E(X^2) = 2\theta_0$ and $E(X^4) = 8\theta_0^2$.

(a) Find the maximum likelihood estimator of θ_0.

(b) Obtain the score function and find the Cramér-Rao Lower Bound for any unbiased estimator of θ_0.

(c) Is your estimator in (a) a UMVUE for θ_0?

(d) Obtain the limiting distribution of your estimator in (a), and state how you would estimate its variance.

(e) Find the maximum likelihood estimator of $\sqrt{\theta_0}$ and derive its asymptotic distribution.

Question 5 (2 parts, 2 points)

Suppose $X \sim N(3,4)$, $Y \sim N(-1,9)$, $Z \sim N(0,25)$ and $W \sim N(2,1)$, mutually independent.

(a) What is the distribution of $\dfrac{9(X-3)^2}{4(Y+1)^2}$?

(b) Using some function of X, Y, Z and W, construct a random variable with a Student t distribution with 3 degrees of freedom.

Question 6 (1 point)

Let $X_1, \ldots, X_n$ be a random sample from a Uniform$(0, \theta_0)$ distribution, for which $E(X) = \theta_0/2$ and $\text{Var}(X) = \theta_0^2/12$. Show that $\widehat{\theta} = 2\overline{X}$ is a consistent estimator of θ_0.

Question 7 (1 point)

Let $X_1, \ldots, X_n$ be a random sample from a Poisson(λ_0) distribution, for which $E(X) = \text{Var}(X) = \lambda_0$ and the MLE of λ_0 is $\overline{X}$. Suppose we wish to estimate $\tau_0 = \lambda_0^2$. Find the maximum likelihood estimator of τ_0 and derive its asymptotic distribution.

Question 8 (1 point)

Suppose $\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, V)$ with $V > 0$, and $\widehat{c} \xrightarrow{p} c$ where $c \neq 0$ is a constant. What is the limiting distribution of $\widehat{c}\sqrt{n}\,(\widehat{\theta} - \theta_0)$?

Question 9 (1 point)

Let $X_1, \ldots, X_n$ be a random sample from a distribution with pdf

$$f(x \mid \theta_0) = \frac{3x^2}{\theta_0^3}, \qquad 0 < x < \theta_0$$

and zero otherwise. Consider $\widehat{\theta} = \frac{4}{3}\overline{X}$. Show that $\widehat{\theta}$ is unbiased, compute $\text{Var}(\widehat{\theta})$, compute the quantity $1/\big(nE[S^2(\theta_0, X)]\big)$, and comment on what you find.

Question 10 (1 point)

Let $X_1, \ldots, X_n$ be a random sample from a Gamma(α_0, β_0) distribution, for which $E(X) = \alpha_0\beta_0$ and $\text{Var}(X) = \alpha_0\beta_0^2$. Find the method of moments estimators of α_0 and β_0.

Mock final solutions

Question 1 (1 point)

SOLUTION

Means.

$$\begin{aligned} E(\overline{X}-\overline{Y}) &= E(\overline{X}) - E(\overline{Y}) && \text{linearity of expectation} \\ &= \frac{1}{n}\sum_{i=1}^{n} E(X_i) - \frac{1}{m}\sum_{i=1}^{m} E(Y_i) && \text{linearity again} \\ &= \frac{n\mu_x}{n} - \frac{m\mu_y}{m} = \mu_x - \mu_y && \text{identically distributed within each sample} \end{aligned}$$

Variances.

$$\begin{aligned} \mathrm{Var}(\overline{X}-\overline{Y}) &= \mathrm{Var}(\overline{X}) + \mathrm{Var}(\overline{Y}) - 2\mathrm{Cov}(\overline{X},\overline{Y}) \\ &= \mathrm{Var}(\overline{X}) + \mathrm{Var}(\overline{Y}) && \textbf{samples independent: } \mathrm{Cov}(X_i, Y_j) = 0 \\ &= \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) + \frac{1}{m^2}\sum_{i=1}^{m}\mathrm{Var}(Y_i) && \textbf{within-sample independence} \\ &= \frac{n\sigma_x^2}{n^2} + \frac{m\sigma_y^2}{m^2} = \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m} \end{aligned}$$

Two independence claims are needed and both must be stated: across the samples (to kill $\mathrm{Cov}(\overline{X},\overline{Y})$) and within each sample (to turn the variance of a sum into a sum of variances). Note also the plus sign in the final answer despite the minus sign in the estimator: variances of independent things always add.

Question 2 (1 point)

SOLUTION

$\widehat{\mu} = \overline{X}$**:** unbiased, so $\mathrm{MSE}(\widehat{\mu}) = \mathrm{Var}(\overline{X}) = \dfrac{\sigma^2}{n} = \dfrac{4}{n}$.
$\widetilde{\mu}$**:** $E(\widetilde{\mu}) = \mu_0 + 0.5$, so Bias $= 0.5$ and $\mathrm{Bias}^2 = 0.25$. Its variance is

$$\mathrm{Var}(\widetilde{\mu}) = \frac{1}{9}\left[\mathrm{Var}(X_1) + \mathrm{Var}(X_2) + \mathrm{Var}(X_3)\right] = \frac{3\cdot 4}{9} = \frac{4}{3}$$

so $\mathrm{MSE}(\widetilde{\mu}) = \dfrac{4}{3} + \dfrac{1}{4} = \dfrac{19}{12} \approx 1.583$.
Compare. $\mathrm{MSE}(\widehat{\mu}) = 4/n$ is below $19/12$ exactly when $n > 48/19 \approx 2.53$, that is, when $n \geq 3$.

$$\boxed{\text{For } n \geq 3,\ \overline{X} \text{ has the lower MSE. For } n = 1 \text{ or } 2,\ \widetilde{\mu} \text{ does.}}$$

The full-credit answer states the condition. Since $\widetilde{\mu}$ uses three observations, the interesting case is $n \geq 3$, where $\overline{X}$ wins and keeps winning by more as n grows: its MSE goes to zero while $\widetilde{\mu}$'s is stuck above 0.25 forever because of the bias.

Question 3 (1 point)

SOLUTION

One parameter, so one equation. First get the population mean:

$$E(X) = \int_0^1 x(\theta+1)x^\theta\,dx = (\theta+1)\int_0^1 x^{\theta+1}\,dx = (\theta+1)\cdot\frac{1}{\theta+2} = \frac{\theta+1}{\theta+2}$$

Set $E(X) = \overline{X}$ and solve:

$$\frac{\theta+1}{\theta+2} = \overline{X} \implies \theta+1 = \overline{X}\theta + 2\overline{X} \implies \theta(1-\overline{X}) = 2\overline{X}-1$$

$$\boxed{\widehat{\theta}_{MOM} = \frac{2\overline{X}-1}{1-\overline{X}}}$$

Question 4 (5 parts, 4 points)

SOLUTION

(a) MLE.

$$\ln f(x \mid \theta) = \ln x - \ln\theta - \frac{x^2}{2\theta}$$

$$L(\theta) = \sum_{i=1}^n \ln X_i - n\ln\theta - \frac{1}{2\theta}\sum_{i=1}^n X_i^2$$

$$\frac{\partial L}{\partial\theta} = -\frac{n}{\theta} + \frac{1}{2\theta^2}\sum_{i=1}^n X_i^2 = 0$$

Multiply by θ^2: $-n\theta + \frac{1}{2}\sum X_i^2 = 0$, so

$$\boxed{\widehat{\theta}_{MLE} = \frac{1}{2n}\sum_{i=1}^n X_i^2}$$

(b) Score and CRLB.

$$S(X,\theta) = \frac{\partial \ln f}{\partial\theta} = -\frac{1}{\theta} + \frac{X^2}{2\theta^2}$$

The score is not linear in X, so differentiate again rather than computing $\mathrm{Var}[S]$:

$$\frac{\partial S}{\partial\theta} = \frac{1}{\theta^2} - \frac{X^2}{\theta^3}$$

$$I(\theta_0) = -E\left[\frac{\partial S}{\partial\theta}\right] = -\left[\frac{1}{\theta_0^2} - \frac{E(X^2)}{\theta_0^3}\right] = -\left[\frac{1}{\theta_0^2} - \frac{2\theta_0}{\theta_0^3}\right] = -\left[\frac{1}{\theta_0^2} - \frac{2}{\theta_0^2}\right] = \frac{1}{\theta_0^2}$$

$$\boxed{\text{CRLB} = \frac{1}{nI(\theta_0)} = \frac{\theta_0^2}{n}}$$

(c) Is $\widehat{\theta}_{MLE}$ a UMVUE? Two checks.

- **Unbiased?** $E(\widehat{\theta}_{MLE}) = \dfrac{1}{2n}\sum E(X_i^2) = \dfrac{1}{2n}\cdot n \cdot 2\theta_0 = \theta_0$. ✓
- **Variance equals the bound?** First, $\text{Var}(X^2) = E(X^4) - [E(X^2)]^2 = 8\theta_0^2 - 4\theta_0^2 = 4\theta_0^2$. Then, since the X_i^2 are i.i.d.,

$$\text{Var}(\widehat{\theta}_{MLE}) = \frac{1}{4n^2}\sum_{i=1}^{n}\text{Var}(X_i^2) = \frac{1}{4n^2}\cdot n \cdot 4\theta_0^2 = \frac{\theta_0^2}{n} = \text{CRLB}\ \checkmark$$

Yes. $\widehat{\theta}_{MLE}$ is unbiased and attains the CRLB, so it is the UMVUE for θ_0.

(d) Limiting distribution and variance estimation. By the asymptotic properties of the MLE, using $I(\theta_0) = 1/\theta_0^2$:

$$\sqrt{n}\left(\widehat{\theta}_{MLE} - \theta_0\right) \xrightarrow{d} N(0,\ \theta_0^2) \qquad \text{so} \qquad \widehat{\theta}_{MLE} \overset{a}{\sim} N\left(\theta_0,\ \frac{\theta_0^2}{n}\right)$$

The asymptotic variance θ_0^2/n depends on the unknown θ_0, so by the invariance property of the MLE, estimate it as

$$\widehat{\text{AVar}}(\widehat{\theta}_{MLE}) = \frac{\widehat{\theta}_{MLE}^2}{n} = \frac{1}{n}\left(\frac{1}{2n}\sum_{i=1}^{n}X_i^2\right)^2$$

(e) MLE of $\sqrt{\theta_0}$ and its asymptotic distribution. By invariance,

$$\widehat{\sqrt{\theta}}_{MLE} = \sqrt{\widehat{\theta}_{MLE}} = \sqrt{\frac{1}{2n}\sum_{i=1}^{n}X_i^2}$$

By the delta method with $g(\theta) = \sqrt{\theta}$, so $g'(\theta) = \frac{1}{2\sqrt{\theta}}$ and $[g'(\theta_0)]^2 = \frac{1}{4\theta_0}$, and $\Lambda = \theta_0^2$ from (d):

$$[g'(\theta_0)]^2\Lambda = \frac{\theta_0^2}{4\theta_0} = \frac{\theta_0}{4}$$

$$\boxed{\sqrt{n}\left(\sqrt{\widehat{\theta}_{MLE}} - \sqrt{\theta_0}\right) \xrightarrow{d} N\left(0,\ \frac{\theta_0}{4}\right)} \qquad \text{and} \qquad \sqrt{\widehat{\theta}_{MLE}} \overset{a}{\sim} N\left(\sqrt{\theta_0},\ \frac{\theta_0}{4n}\right)$$

Question 5 (2 parts, 2 points)

SOLUTION

(a) Regroup so each squared deviation sits over its own variance. Here $\sigma_x^2 = 4$ and $\sigma_y^2 = 9$:

$$\frac{9(X-3)^2}{4(Y+1)^2} = \frac{(X-3)^2/4}{(Y+1)^2/9} = \frac{\left(\dfrac{X-3}{2}\right)^2}{\left(\dfrac{Y-(-1)}{3}\right)^2} = \frac{\chi^2_{(1)}/1}{\chi^2_{(1)}/1}$$

Both are squared standard normals, hence $\chi^2_{(1)}$, and they are independent because X and Y are.

Therefore

$$\boxed{\sim F_{(1,1)}}$$

(b) A $t_{(3)}$ needs one standard normal on top and an independent $\chi^2_{(3)}$ on the bottom, divided by 3. Standardize each variable:

$$\frac{X-3}{2}, \quad \frac{Y+1}{3}, \quad \frac{Z}{5}, \quad \frac{W-2}{1} \text{ are each } N(0,1), \text{ mutually independent}$$

Spend one on top and the other three on the bottom:

$$V = \left(\frac{Y+1}{3}\right)^2 + \left(\frac{Z}{5}\right)^2 + (W-2)^2 \sim \chi^2_{(3)}$$

as a sum of three independent squared standard normals. Since X is independent of Y, Z, W, it is independent of V, so

$$\boxed{\frac{\frac{X-3}{2}}{\sqrt{V/3}} \sim t_{(3)}}$$

Question 6 (1 point)

SOLUTION

Bias. $E(\widehat{\theta}) = 2E(\overline{X}) = 2E(X) = 2 \cdot \frac{\theta_0}{2} = \theta_0$, so $\text{Bias}(\widehat{\theta}) = 0$ and trivially $\lim_{n\to\infty} \text{Bias}(\widehat{\theta}) = 0$.
Variance.

$$\text{Var}(\widehat{\theta}) = \text{Var}(2\overline{X}) = 4\text{Var}(\overline{X}) = \frac{4\text{Var}(X)}{n} = \frac{4}{n} \cdot \frac{\theta_0^2}{12} = \frac{\theta_0^2}{3n}$$

$$\lim_{n\to\infty} \text{Var}(\widehat{\theta}) = \lim_{n\to\infty} \frac{\theta_0^2}{3n} = 0$$

Conclusion. Since $\lim \text{Bias}(\widehat{\theta}) = 0$ and $\lim \text{Var}(\widehat{\theta}) = 0$, we have $\widehat{\theta} \xrightarrow{q.m.} \theta_0$. Because convergence in mean square implies convergence in probability, $\widehat{\theta} \xrightarrow{p} \theta_0$, and therefore $\widehat{\theta} = 2\overline{X}$ is a consistent estimator of θ_0. ■
Alternative one-line route: by the WLLN, $\overline{X} \xrightarrow{p} E(X) = \theta_0/2$, and by continuous mapping with $g(u) = 2u$, $2\overline{X} \xrightarrow{p} \theta_0$. Both are acceptable; the two-limit version is what he modeled in the PS10 key, so it is the safer answer to write.

Question 7 (1 point)

SOLUTION

The estimator, by invariance. Since $\widehat{\lambda}_{MLE} = \overline{X}$ and $\tau_0 = g(\lambda_0) = \lambda_0^2$,

$$\widehat{\tau}_{MLE} = \left(\widehat{\lambda}_{MLE}\right)^2 = \overline{X}^2$$

The limiting distribution of $\widehat{\lambda}_{MLE}$ first. The score is $S(X,\lambda) = \frac{X}{\lambda} - 1$, so

$$I(\theta_0) = \text{Var}\left[\frac{X}{\lambda_0} - 1\right] = \frac{\text{Var}(X)}{\lambda_0^2} = \frac{\lambda_0}{\lambda_0^2} = \frac{1}{\lambda_0} \quad \Longrightarrow \quad \sqrt{n}(\overline{X} - \lambda_0) \xrightarrow{d} N(0, \lambda_0)$$

Now the delta method with $g(\lambda) = \lambda^2$, so $g'(\lambda) = 2\lambda$ and $[g'(\lambda_0)]^2 = 4\lambda_0^2$:

$$[g'(\lambda_0)]^2 \Lambda = 4\lambda_0^2 \cdot \lambda_0 = 4\lambda_0^3$$

$$\boxed{\sqrt{n}\left(\overline{X}^2 - \lambda_0^2\right) \xrightarrow{d} N(0,\ 4\lambda_0^3)} \qquad \text{and} \qquad \overline{X}^2 \overset{a}{\sim} N\left(\lambda_0^2,\ \frac{4\lambda_0^3}{n}\right)$$

Question 8 (1 point)

SOLUTION

One sequence converges in distribution, the other in probability to a constant, and they are being multiplied. That is **Slutsky's theorem**, part (i): $X_n Y_n \xrightarrow{d} aU$.
Here $U \sim N(0, V)$ and $a = c$, so

$$\widehat{c}\,\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} c \cdot N(0, V)$$

Multiplying a normal by a constant multiplies its variance by the constant squared:

$$\boxed{\widehat{c}\,\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0,\ c^2 V)}$$

Note that $\widehat{c}$ is replaced by its probability limit c, not left as $\widehat{c}$. The whole point of Slutsky is that a consistently estimated constant behaves asymptotically like the constant itself.

Question 9 (1 point)

SOLUTION

Unbiasedness.

$$E(X) = \int_0^{\theta_0} x \cdot \frac{3x^2}{\theta_0^3}\,dx = \frac{3}{\theta_0^3} \cdot \frac{\theta_0^4}{4} = \frac{3\theta_0}{4} \quad \Longrightarrow \quad E(\widehat{\theta}) = \frac{4}{3} \cdot \frac{3\theta_0}{4} = \theta_0 \ \checkmark$$

Variance.

$$E(X^2) = \int_0^{\theta_0} x^2 \cdot \frac{3x^2}{\theta_0^3}\,dx = \frac{3}{\theta_0^3} \cdot \frac{\theta_0^5}{5} = \frac{3\theta_0^2}{5}$$

$$\text{Var}(X) = \frac{3\theta_0^2}{5} - \left(\frac{3\theta_0}{4}\right)^2 = \frac{3\theta_0^2}{5} - \frac{9\theta_0^2}{16} = \frac{48\theta_0^2 - 45\theta_0^2}{80} = \frac{3\theta_0^2}{80}$$

$$\text{Var}(\widehat{\theta}) = \frac{16}{9} \cdot \frac{\text{Var}(X)}{n} = \frac{16}{9} \cdot \frac{3\theta_0^2}{80n} = \frac{\theta_0^2}{15n}$$

The bound-like quantity.

$$\ln f(x \mid \theta) = \ln 3 + 2\ln x - 3\ln\theta \quad \Longrightarrow \quad S(X,\theta) = -\frac{3}{\theta}$$

$$E\left[S^2(\theta_0, X)\right] = \frac{9}{\theta_0^2} \quad \Longrightarrow \quad \frac{1}{nE[S^2]} = \frac{\theta_0^2}{9n}$$

Comment, which is the actual point of the question.

$$\mathrm{Var}(\widehat{\theta}) = \frac{\theta_0^2}{15n} < \frac{\theta_0^2}{9n}$$

An unbiased estimator has variance strictly *below* the apparent Cramér-Rao Lower Bound. This is not a contradiction. **The support of X is $(0, \theta_0)$, which depends on the parameter**, so the regularity conditions of the Cramér-Rao theorem fail: one cannot interchange differentiation and integration when the limits of integration move with θ. The bound therefore does not apply here, and the three formulas for $I(\theta_0)$ need not agree.
Corroborating tell: the score $S = -3/\theta$ does not involve X at all, so $E[S(\theta_0, X)] = -3/\theta_0 \neq 0$ and the **score identity itself fails**. That is the cleanest single sentence you can write to show you understand what has gone wrong.

Question 10 (1 point)

SOLUTION

Two parameters, so two equations, using centered moments:

$$E(X) = \alpha\beta = \overline{X} \qquad \text{and} \qquad \mathrm{Var}(X) = \alpha\beta^2 = \widetilde{S}^2$$

Divide the second equation by the first to eliminate α:

$$\frac{\alpha\beta^2}{\alpha\beta} = \beta = \frac{\widetilde{S}^2}{\overline{X}}$$

Substitute back into the first:

$$\alpha = \frac{\overline{X}}{\beta} = \frac{\overline{X}}{\widetilde{S}^2/\overline{X}} = \frac{\overline{X}^2}{\widetilde{S}^2}$$

$$\boxed{\widehat{\alpha}_{MOM} = \frac{\overline{X}^2}{\widetilde{S}^2}} \qquad \boxed{\widehat{\beta}_{MOM} = \frac{\widetilde{S}^2}{\overline{X}}}$$

Worth a sentence: the Gamma MLE has no closed form and would require numerical maximization, which is precisely why MOM is the right tool here.

MEMORIZE

Grading yourself. Thirteen scored parts. Award full credit only if you named the theorem where one applies and stated independence where you used it.

- **11 or more:** you are at or above the target. Spend remaining time transcribing the cheat sheets by hand.
- **8 to 10:** identify which machine failed and reread that block only. The most common gaps are Q4b (choosing the right information formula) and Q9 (recognizing the support trap).
- **7 or fewer:** reread Block 5, then redo Sample Final Q4 from a blank page, then retake this.

CHEAT SHEET 1 • SIDE A

THE SECOND HALF: ESTIMATION AND ASYMPTOTICS

ordered by exam probability, highest first

Formulas and procedural steps only. θ_0 true value (fixed) • θ any candidate • $\widehat{\theta}$ estimator (random) • $\overline{X} = \frac{1}{n}\sum X_i$ • $S^2 = \frac{1}{n-1}\sum(X_i - \overline{X})^2$ • $\widetilde{S}^2 = \frac{1}{n}\sum(X_i - \overline{X})^2 = \frac{n-1}{n}S^2$

0. WHICH MACHINE DOES THIS QUESTION WANT?

Trigger phrase	Go to
"find the MOM estimator"	box 3
"find the MLE"	box 2
"find the CRLB"	box 1, steps 0–3
"is it a UMVUE?"	box 5
"limiting distribution of"	box 1, step 3
"asymptotic / approximate dist."	box 1, step 4
"how would you estimate its variance"	box 1, step 4
"estimate $g(\theta_0)$ and its distribution"	box 4
"show it is consistent"	box 6
"which has lower MSE"	box 10
"consistent, asy. normal, asy. efficient"	the MLE, box 11
"construct a χ^2 / t / F"	box 8
"prove $E(\overline{X})$, $\mathrm{Var}(\overline{X})$, $E(S^2)$"	box 9
"verify the score / info identity"	box 12
two sequences, one $\xrightarrow{d}$ one $\xrightarrow{p}$	Slutsky, box 11
one estimator through a function	delta, box 4
support of x contains θ	box 13 first

1. THE MASTER PIPELINE (answers four question types at once)

0. $\ln f(x \mid \theta)$ ONE observation, expand fully

1. $S(\theta, X) = \dfrac{\partial \ln f(X \mid \theta)}{\partial \theta}$ *no summation sign*

2. $I(\theta_0) = -E\left[\dfrac{\partial S(\theta_0, X)}{\partial \theta}\right] = \mathrm{Var}[S(\theta_0, X)] = E[S^2(\theta_0, X)]$

3. CRLB $= \dfrac{1}{nI(\theta_0)}$ and $\sqrt{n}(\widehat{\theta}_{MLE} - \theta_0) \xrightarrow{d} N\left(0, [I(\theta_0)]^{-1}\right)$

4. $\widehat{\theta}_{MLE} \overset{a}{\sim} N\left(\theta_0, \dfrac{1}{nI(\theta_0)}\right)$; $\widehat{\mathrm{AVar}} = \dfrac{1}{nI(\theta)}\Big|_{\theta=\widehat{\theta}}$

Choosing the $I(\theta_0)$ formula. Score linear in X ⇒ $\mathrm{Var}[S]$, one line. Score nonlinear ⇒ differentiate again, $-E[\partial S/\partial\theta]$. Regularity fails ⇒ use the definition $E[S^2]$.

Key fact: AVar of the MLE = CRLB. That is asymptotic efficiency.

$I(\theta_0)$ is per observation. If an n appears inside your $I(\theta_0)$, you summed the score by mistake.

2. MLE STEPS

1. Log the density for one observation, expand fully.

2. $L(\theta) = \sum_{i=1}^{n} \ln f(X_i \mid \theta)$; terms free of θ become constants.

3. $\partial L/\partial\theta = 0$ (first-order condition).

4. Solve for θ; write $\widehat{\theta}_{MLE}$ with capital X's.

- Second-order conditions are not required.
- Parameter σ^2: differentiate with respect to σ^2, not σ. Set $v = \sigma^2$ and work in v.
- Support depends on θ: calculus fails. Impose $\theta \geq \max\{X_i\}$, then use monotonicity of $-n\ln\theta$.
- Absolute value in the density: not differentiable; the MLE is the sample median.
- No closed form (two-parameter Gamma): say numerical maximization is required, and use MOM instead.

Invariance. MLE of $g(\theta_0)$ is $g(\widehat{\theta}_{MLE})$. Plug in, no new work. Used for $\sqrt{\theta}$, $1/\theta$, θ^2, $e^{-\theta}$, and for $\widehat{\mathrm{AVar}}$.

3. MOM STEPS

1. Count parameters k; you need k equations.

2. Write population moments as functions of θ; integrate if not given.

3. Set equal to sample counterparts, *centered*:

$$E(X) = \overline{X}, \qquad \mathrm{Var}(X) = \widetilde{S}^2$$

4. Solve the system; label $\widehat{\theta}_{MOM}$.

- NEVER pair $\mathrm{Var}(X)$ with $\frac{1}{n}\sum X_i^2$, nor $\widetilde{S}^2$ with $E(X^2)$.
- Uncentered alternative: $E(X) = \overline{X}$ and $E(X^2) = \frac{1}{n}\sum X_i^2$.
- The sample second moment uses $\frac{1}{n}$, not $\frac{1}{n-1}$.
- If both equations are products of powers of the parameters, divide one by the other to eliminate one parameter.
- If the parameter is a standard deviation, take a square root at the end.
- Drawbacks: can fall outside Θ, may be inefficient. Virtue: almost always closed form.

4. DELTA METHOD STEPS (estimating $\gamma_0 = g(\theta_0)$)

1. Estimator by invariance: $\widehat{\gamma} = g(\widehat{\theta})$.

2. Compute $g'(\theta)$, square it, evaluate at θ_0.

3. Multiply by Λ from $\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, \Lambda)$. For an MLE, $\Lambda = [I(\theta_0)]^{-1}$.

4. Write both forms:

$$\sqrt{n}\left[g(\widehat{\theta}) - g(\theta_0)\right] \xrightarrow{d} N\left(0, [g'(\theta_0)]^2\Lambda\right)$$

$$g(\widehat{\theta}) \overset{a}{\sim} N\left(g(\theta_0),\ \tfrac{1}{n}[g'(\theta_0)]^2\Lambda\right)$$

Conditions: g continuously differentiable, $g'(\theta_0) \neq 0$. Center at $g(\theta_0)$, never at θ_0. Square the derivative; it enters the variance slot.

Squared derivatives you may need:

$g = \sqrt{\theta} \Rightarrow [g']^2 = \frac{1}{4\theta}$ $\quad$ $g = 1/\theta \Rightarrow [g']^2 = \theta^{-4}$

$g = \theta^2 \Rightarrow [g']^2 = 4\theta^2$ $\quad$ $g = e^{-\theta} \Rightarrow [g']^2 = e^{-2\theta}$

$g = \ln\theta \Rightarrow [g']^2 = \theta^{-2}$ $\quad$ $g = \theta^k \Rightarrow [g']^2 = k^2\theta^{2k-2}$

5. UMVUE DECISION STEPS

1. Unbiased? If $E(\widehat{\theta}) \neq \theta_0$, STOP. Answer "No: biased estimators cannot be UMVUE." Nothing further is needed.

2. Compute $\mathrm{Var}(\widehat{\theta})$ and compare with CRLB $= 1/(nI(\theta_0))$.

3. Unbiased AND Var = CRLB ⇒ it is the UMVUE, and no unbiased estimator has smaller variance.

- Converse is false: a UMVUE need not attain the bound, so Var > CRLB does not prove "not UMVUE."
- The CRLB bounds the VARIANCE of UNBIASED estimators only. A biased estimator may sit below it legitimately.
- Definition: (i) $E(\widehat{\theta}^*) = \theta_0$ for all $\theta_0 \in \Theta$; (ii) $\mathrm{Var}(\widehat{\theta}^*) \leq \mathrm{Var}(\widehat{\theta})$ for every other unbiased $\widehat{\theta}$ and every θ_0. "Uniform" means at all θ_0.
- Restricting to unbiased estimators is what rules out constant estimators.

6. CONSISTENCY PROOF STEPS

1. Compute $E(\widehat{\theta})$, hence Bias $= E(\widehat{\theta}) - \theta_0$; show $\lim_{n\to\infty}$ Bias $= 0$.

2. Compute $\mathrm{Var}(\widehat{\theta})$; show $\lim_{n\to\infty}\mathrm{Var}(\widehat{\theta}) = 0$.

3. Conclude in these words: both limits are zero, so $\widehat{\theta} \xrightarrow{q.m.} \theta_0$; mean-square convergence implies convergence in probability, so $\widehat{\theta} \xrightarrow{p} \theta_0$; therefore $\widehat{\theta}$ is consistent for θ_0.

Shortcuts. WLLN gives $\overline{X} \xrightarrow{p} \mu$ in one line. Continuous mapping: $\widehat{\theta} \xrightarrow{p} \theta_0$ and g continuous ⇒ $g(\widehat{\theta}) \xrightarrow{p} g(\theta_0)$, i.e. $\mathrm{plim}\, g(\widehat{\theta}) = g(\mathrm{plim}\,\widehat{\theta})$.

Definition: $\lim_{n\to\infty} P(|\widehat{\theta} - \theta_0| > \epsilon) = 0$ for every $\epsilon > 0$.

Biased and consistent are compatible; asymptotic unbiasedness suffices.

Converse fails: consistency does not require the variance to go to zero.

7. VARIANCE AND EXPECTATION EXTRACTION

$E(a + bY) = a + bE(Y)$, $\quad \mathrm{Var}(a + bY) = b^2\mathrm{Var}(Y)$

$\mathrm{Var}(Y) = E(Y^2) - [E(Y)]^2 \quad \Leftrightarrow \quad E(Y^2) = \mathrm{Var}(Y) + [E(Y)]^2$

If $cY \sim \chi^2_{(r)}$: $E(Y) = r/c$ and $\mathrm{Var}(Y) = 2r/c^2$

$\mathrm{Var}(\sum_{i=1}^{n} Y_i) = \sum_i \mathrm{Var}(Y_i)$ *only under independence*

$\mathrm{Var}(aX + bY) = a^2\mathrm{Var}X + b^2\mathrm{Var}Y + 2ab\mathrm{Cov}(X, Y)$

$\mathrm{Var}(\overline{X}) = \mathrm{Var}(X)/n$; $\mathrm{Var}(c\overline{X}) = c^2\mathrm{Var}(X)/n$

If $\widehat{\theta} = c\overline{X}$: $E(\widehat{\theta}) = c\,E(X)$, $\mathrm{Var}(\widehat{\theta}) = c^2\mathrm{Var}(X)/n$

If $\widehat{\theta} = \frac{1}{cn}\sum h(X_i)$: $E(\widehat{\theta}) = \frac{E[h(X)]}{c}$, $\mathrm{Var}(\widehat{\theta}) = \frac{\mathrm{Var}[h(X)]}{c^2 n}$

8. PROOF SKELETONS (annotate every line; the annotations are the grade)

$E(\overline{X}) = \mu$**.** definition of $\overline{X}$ → linearity, pull out $\frac{1}{n}$ → expectation of a sum is the sum of expectations → *identically distributed*, so each $E(X_i) = \mu$ → $\frac{1}{n}\cdot n\mu$.

$\mathrm{Var}(\overline{X}) = \sigma^2/n$**.** definition → $\mathrm{Var}(aY) = a^2\mathrm{Var}(Y)$, so $\frac{1}{n}$ becomes $\frac{1}{n^2}$ → **by independence**, $\mathrm{Cov}(X_i, X_j) = 0$, so the variance of the sum is the sum of variances → identically distributed → $\frac{1}{n^2}n\sigma^2$.

$E(S^2) = \sigma^2$**.** start from $(n-1)S^2 = \sum X_i^2 - n\overline{X}^2$ → take expectations → apply the flip to both pieces, $E(X_i^2) = \sigma^2 + \mu^2$ and $E(\overline{X}^2) = \frac{\sigma^2}{n} + \mu^2$ → the μ^2 terms cancel and $n-1$ copies of σ^2 remain.

Two independent samples. same two chains, plus one extra sentence killing $\mathrm{Cov}(\overline{X}, \overline{Y}) = 0$ across samples before splitting the variance.

Named justifications to write: linearity of expectation; $\mathrm{Var}(aY) = a^2\mathrm{Var}(Y)$; by independence; identically distributed; Thm 5.2.6; Thm 5.3.1(c).

8. BUILD A χ^2, t, OR F

1. Count what you need. $t_{(p)}$: one $N(0,1)$ plus an independent $\chi^2_{(p)}$. $F_{(r_1,r_2)}$: two independent chi-squares.

2. Standardize every normal: $Z_j = \dfrac{X_j - \mu_j}{\sigma_j} \sim N(0,1)$.

Divide by the standard *deviation*.

3. Build chi-squares by squaring and summing as many Z_j as you need degrees of freedom.

4. Form the ratio, dividing each χ^2 by its OWN df, and state which variables are independent of which.

Never reuse a variable in numerator and denominator: independence would fail. Allocate one variable to the top, the rest to the bottom.

Building blocks.

$Z \sim N(0,1) \Rightarrow Z^2 \sim \chi^2_{(1)}$

$Y_i \sim \chi^2_{(p_i)}$ independent $\Rightarrow \sum Y_i \sim \chi^2_{(\sum p_i)}$

m independent $N(0,1)$, squared and summed $\Rightarrow \chi^2_{(m)}$

$U \sim N(0,1)$, $V \sim \chi^2_{(p)}$, independent:

$$\frac{U}{\sqrt{V/p}} \sim t_{(p)}$$

$U \sim \chi^2_{(r_1)}$, $V \sim \chi^2_{(r_2)}$, independent:

$$\frac{U/r_1}{V/r_2} \sim F_{(r_1,r_2)}$$

Shapes: $t = \dfrac{\text{normal}}{\sqrt{\chi^2/\text{its own df}}}$, $\quad F = \dfrac{\chi^2/\text{its own df}}{\chi^2/\text{its own df}}$

The t has a square root over the whole denominator; the F does not.

To simplify a messy ratio, regroup so each squared deviation sits over its own variance.

9. SAMPLING DISTRIBUTION FACTS

ANY f_X with mean μ, variance σ^2 (Thm 5.2.6). No normality needed.

$$E(\overline{X}) = \mu, \quad \text{Var}(\overline{X}) = \frac{\sigma^2}{n}, \quad E(S^2) = \sigma^2$$

NORMAL ONLY (Thm 5.3.1).

$$\overline{X} \sim N\left(\mu, \tfrac{\sigma^2}{n}\right); \quad \overline{X} \perp S^2; \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

Consequences under normality.

$$\frac{\sqrt{n}(\overline{X}-\mu)}{\sigma} \sim N(0,1), \quad \frac{\sqrt{n}(\overline{X}-\mu)}{S} \sim t_{(n-1)}$$

$$\frac{\sum(X_i-\mu)^2}{\sigma^2} \sim \chi^2_{(n)}, \quad \frac{n(\overline{X}-\mu)^2}{\sigma^2} \sim \chi^2_{(1)}$$

$$\text{Var}(S^2) = \frac{2\sigma^4}{n-1}, \qquad \text{Var}(\widetilde{S}^2) = \frac{2(n-1)\sigma^4}{n^2}$$

$$E(\widetilde{S}^2) = \frac{n-1}{n}\sigma^2, \qquad \text{Bias}(\widetilde{S}^2) = -\frac{\sigma^2}{n}$$

Mean known to be 0: $\frac{1}{n}\sum X_i^2$ is unbiased for σ^2, with variance $\frac{2\sigma^4}{n}$.

Degrees of freedom rule: subtracting the known μ gives n; subtracting the estimated $\overline{X}$ costs one and gives $n-1$.

9b. TWO SAMPLES, AND FOUR IDENTITIES

$$E(\overline{X}-\overline{Y}) = \mu_x - \mu_y, \quad \text{Var}(\overline{X}-\overline{Y}) = \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}$$

(plus sign: independent variances add)

$$\frac{s_x^2/\sigma_x^2}{s_y^2/\sigma_y^2} \sim F_{(n-1,m-1)}$$

$\text{Cov}(X_i - \overline{X}, \overline{X}) = 0$; $\sum(X_i - \overline{X}) = 0$;

$(n-1)S^2 = \sum X_i^2 - n\overline{X}^2$; $M_{\overline{X}}(t) = [M_X(t/n)]^n$

10. MSE AND BIAS

$$\text{MSE}(\widehat{\theta}) = E[(\widehat{\theta}-\theta_0)^2] = \text{Var}(\widehat{\theta}) + \text{Bias}^2(\widehat{\theta})$$

$$\text{Bias}(\widehat{\theta}) = E(\widehat{\theta}) - \theta_0$$

Unbiased $\Rightarrow$ MSE = Var: half of all MSE questions are variance questions.

Derivation if asked: put $Y = \widehat{\theta} - \theta_0$ into $E(Y^2) = \text{Var}Y + [EY]^2$, and note $\text{Var}(\widehat{\theta} - \theta_0) = \text{Var}(\widehat{\theta})$ since θ_0 is a constant.

Comparison steps: for each estimator get E, then bias, then Var, then add, then compare. **If the ranking depends on n or on θ_0, state the condition explicitly.**

Variance shrinks with n; a fixed bias does not, so a non-vanishing bias eventually dominates.

11. ASYMPTOTIC THEOREMS

Chain: $\xrightarrow{q.m.} \Rightarrow \xrightarrow{p} \Rightarrow \xrightarrow{d}$. No reverses.

$X_n \xrightarrow{q.m.} c$: $\lim E[(X_n - c)^2] = 0$

$X_n \xrightarrow{p} c$: $\lim P(|X_n - c| > \epsilon) = 0$ for all $\epsilon > 0$

$X_n \xrightarrow{d} Y$: $\lim F_{X_n}(r) = F_Y(r)$

WLLN: $\overline{X} \xrightarrow{p} \mu$. *Proof:* Chebyshev gives $P(|\overline{X} - \mu| > \epsilon) \le \frac{\text{Var}(\overline{X})}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \to 0$.

CLT (Lindeberg-Lévy): for any f_X with mean μ and variance σ^2,

$$\sqrt{n}(\overline{X}-\mu) \xrightarrow{d} N(0,\sigma^2), \quad \frac{\sqrt{n}(\overline{X}-\mu)}{\sigma} \xrightarrow{d} N(0,1)$$

$$\Longrightarrow \quad \overline{X} \overset{a}{\sim} N\left(\mu, \tfrac{\sigma^2}{n}\right)$$

The CLT is about $\overline{X}$, not the individual X_i. The $\sqrt{n}$ is the unique rate that stabilizes the variance.

11b. SLUTSKY, EFFICIENCY, THE MLE THEOREM

Probability limit algebra: $X_n \xrightarrow{p} a$, $Y_n \xrightarrow{p} b \Rightarrow X_n + Y_n \xrightarrow{p} a+b$; $X_n Y_n \xrightarrow{p} ab$; $X_n/Y_n \xrightarrow{p} a/b$ if $b \neq 0$.

Slutsky: $X_n \xrightarrow{d} U$ (random), $Y_n \xrightarrow{p} a$ (constant):

$$X_n Y_n \xrightarrow{d} aU, \quad X_n + Y_n \xrightarrow{d} U + a, \quad \frac{X_n}{Y_n} \xrightarrow{d} \frac{U}{a} \ (a \neq 0)$$

Pulling a constant into a normal squares it: $a \cdot N(0,V) = N(0, a^2 V)$.

Asymptotic efficiency: $\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, [I(\theta_0)]^{-1})$, i.e. asymptotically unbiased with AVar = CRLB.

MLE theorem. $\widehat{\theta}_{MLE}$ is (a) consistent, (b) asymptotically normal, $\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, [I(\theta_0)]^{-1})$, (c) asymptotically efficient, $\widehat{\theta} \overset{a}{\sim} N(\theta_0, [nI(\theta_0)]^{-1})$.

The MLE is usually *biased* in finite samples.

12. THE TWO IDENTITIES

Score identity: $E[S(\theta_0, X)] = 0$

Information matrix identity:

$$-E\left[\frac{\partial S(\theta_0, X)}{\partial \theta}\right] = E\left[S^2(\theta_0, X)\right]$$

Both hold ONLY at θ_0. At a general θ they fail, and that failure is usually the point of the question.

Since $E[S] = 0$, $\text{Var}[S] = E[S^2]$, which is why two of the three $I(\theta_0)$ formulas coincide.

$E[(X-\mu)^2]$ equals $\text{Var}(X)$ only at the true μ.

Verification steps: substitute θ_0, take the expectation, quote the true $E(X)$, $E(X^2)$ or $E(X^4)$, and show it collapses to 0 or to equality.

Origin: differentiate $\int f(x \mid \theta)\,dx = 1$ once (score) or twice (information) with respect to θ.

13. WHEN THE THEOREMS FAIL

Detector: does the range of x contain θ? Tells: support $(0,\theta_0)$, Uniform$(0,\theta_0)$, an indicator $\mathbf{1}\{x \le \theta\}$, or a score with no X in it.

Then differentiation and integration cannot be interchanged, and:

- the CRLB need not bind, so an unbiased estimator can have variance strictly below it;
- the score identity can fail, $E[S] \neq 0$;
- the three $I(\theta_0)$ formulas can disagree, so use the definition $1/(nE[S^2])$;
- the MLE cannot be found by setting a derivative to zero.

Write the sentence: "the support depends on θ_0, so the regularity conditions fail and the theorem does not apply."

14. NICHE BUT USEFUL

$E(\chi^2_{(r)}) = r$, $\text{Var}(\chi^2_{(r)}) = 2r$

$X \sim F_{(p,q)} \Rightarrow 1/X \sim F_{(q,p)}$

$X \sim t_{(q)} \Rightarrow X^2 \sim F_{(1,q)}$, hence $\frac{n(\overline{X}-\mu)^2}{S^2} \sim F_{(1,n-1)}$

$X \sim N(0,\sigma^2)$: $E(X^2) = \sigma^2$, $E(X^4) = 3\sigma^4$, odd moments zero.

$E(S^2) = \sigma^2$ does NOT give $E(S) = \sigma$; by Jensen, $E(S) < \sigma$.

$E[g(\widehat{\theta})] \neq g[E(\widehat{\theta})]$, but $\text{plim}\, g(\widehat{\theta}) = g(\text{plim}\, \widehat{\theta})$.

Exponential: $\frac{1}{\theta}e^{-x/\theta}$ has $E = \theta$ and MLE $\overline{X}$; $\theta e^{-\theta x}$ has $E = 1/\theta$ and MLE $1/\overline{X}$. Read which one you are given.

Geometric $p(1-p)^{x-1}$: $E(X) = 1/p$, so $\overline{X}$ is biased for p and $\widetilde{p}_{MOM} = 1/\overline{X}$.

$\widehat{\theta}_{MOM} = \widehat{\theta}_{MLE}$ for Normal, Poisson, Bernoulli and exponential-in-mean; not in general.

Random sample = independent + identically distributed; joint density $\prod_i f(x_i)$. A statistic is any function of the sample, hence random.

GMM: more moment conditions than parameters, model over-identified, minimize the distance between sample and population moments.

CHEAT SHEET 2 • SIDE A

THE PROBABILITY TOOLKIT (trimmed midterm sheet)

the first-half tools the final actually uses

Everything here is a tool the estimation questions call on. Set theory, Kolmogorov's axioms, Bayes' rule and the joint-distribution machinery are compressed to Side B, since they are unlikely to carry a question of their own.

1. EXPECTATION AND VARIANCE ALGEBRA (used in almost every answe

$E(aX+b) = aE(X)+b$
$E(X+Y) = E(X)+E(Y)$ *always, no independence needed*
$E(aX+bY) = aE(X)+bE(Y)$
$E(c) = c$ and $\mathrm{Var}(c) = 0$ for a constant c
$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$ *shifts do not change variance*
$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$
$\boxed{E(X^2) = \mathrm{Var}(X) + [E(X)]^2}$ *the flip; use it whenever an* $E(\cdot^2)$ *blocks you*
$\mathrm{Var}(aX+bY) = a^2\mathrm{Var}X + b^2\mathrm{Var}Y + 2ab\mathrm{Cov}(X,Y)$
$\mathrm{Var}(X-Y) = \mathrm{Var}X + \mathrm{Var}Y - 2\mathrm{Cov}(X,Y)$
$\mathrm{Var}(\sum_i X_i) = \sum_i \mathrm{Var}(X_i)$ *iff all covariances vanish*
$E[g(X)] \neq g[E(X)]$ in general.

2. COMPUTING MOMENTS FROM A DENSITY

$$E(X) = \int_{\text{supp}} x f(x)\,dx, \quad E(X^2) = \int_{\text{supp}} x^2 f(x)\,dx$$

$$E[g(X)] = \int_{\text{supp}} g(x) f(x)\,dx \quad \text{(LOTUS)}$$

Discrete: replace $\int$ by $\sum_x$ and f by the pmf.
Always integrate over the stated support, not over $\mathbb{R}$.
Sanity check: $\int_{\text{supp}} f(x)\,dx = 1$.

Integrals that keep appearing:
$\int_0^\theta x^k dx = \dfrac{\theta^{k+1}}{k+1}$ $\quad$ $\int_0^1 x^k dx = \dfrac{1}{k+1}$
$\int_0^\infty x^k e^{-\lambda x} dx = \dfrac{k!}{\lambda^{k+1}}$ (integer $k \geq 0$)
$\int_0^\infty x^{\alpha-1} e^{-x/\beta} dx = \Gamma(\alpha)\beta^\alpha$
An odd function integrated over a symmetric interval gives zero.

3. COVARIANCE AND CORRELATION

$\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y)$
$\mathrm{Cov}(X,X) = \mathrm{Var}(X)$; $\mathrm{Cov}(X,c) = 0$
$\mathrm{Cov}(aX+b,\ cY+d) = ac\,\mathrm{Cov}(X,Y)$
$\mathrm{Cov}(X+W, Y+Z) = \mathrm{Cov}(X,Y) + \mathrm{Cov}(X,Z) + \mathrm{Cov}(W,Y) + \mathrm{Cov}(W,Z)$
$\mathrm{Corr}(X,Y) = \rho = \dfrac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} \in [-1,1]$
Independence $\Rightarrow$ Cov $= 0$ and $E(XY) = E(X)E(Y)$. The converse is false.

4. TRANSFORMATIONS OF RANDOM VARIABLES

CDF method (always works).
1. $F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$
2. Rearrange to isolate X; flip the inequality if you divide by a negative.
3. Write it as $F_X(\cdot)$, then differentiate to get $f_Y(y)$.

Change of variable (monotone g).

$$f_Y(y) = f_X\big(g^{-1}(y)\big)\left|\frac{d}{dy} g^{-1}(y)\right|$$

Find the new support by pushing the endpoints of the old support through g.

Bivariate. $f_{UV}(u,v) = f_{XY}\big(x(u,v), y(u,v)\big)\,|J|$ with

$$J = \det\begin{pmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \end{pmatrix}$$

then integrate out to get a marginal.

Normal facts. $X \sim N(\mu,\sigma^2) \Rightarrow \frac{X-\mu}{\sigma} \sim N(0,1)$; $aX+b \sim N(a\mu+b, a^2\sigma^2)$; sums of independent normals are normal with means and variances adding.

5. MOMENT GENERATING FUNCTIONS

$M_X(t) = E(e^{tX})$
$E(X^k) = M_X^{(k)}(0)$, the kth derivative at zero
$M_{aX+b}(t) = e^{bt} M_X(at)$
$X \perp Y \Rightarrow M_{X+Y}(t) = M_X(t) M_Y(t)$
Uniqueness: the same mgf means the same distribution. This is how you identify a distribution from its mgf.
$M_{\overline{X}}(t) = [M_X(t/n)]^n$
$N(\mu,\sigma^2)$: $M(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$

6. JOINT, MARGINAL, CONDITIONAL

Marginal: $f_X(x) = \int f_{XY}(x,y)\,dy$
Conditional: $f_{Y|X}(y \mid x) = \dfrac{f_{XY}(x,y)}{f_X(x)}$

Independence test (both parts required).
(i) $f_{XY}(x,y) = f_X(x) f_Y(y)$, and
(ii) the support is a **rectangle**: the range of y must not depend on x.
Factorization alone is not sufficient.

$E(Y \mid X = x)$ is a *number*; $E(Y \mid X)$ is a *random variable*.
Law of iterated expectations: $E[E(Y \mid X)] = E(Y)$
Variance decomposition: $\mathrm{Var}(Y) = E[\mathrm{Var}(Y \mid X)] + \mathrm{Var}[E(Y \mid X)]$
$E[g(X)Y \mid X] = g(X)\,E[Y \mid X]$ (pull out what is known)

7. THE THREE INEQUALITIES

Markov: $X \geq 0 \Rightarrow P(X \geq a) \leq \dfrac{E(X)}{a}$

Chebyshev: $P(|X-\mu| \geq k\sigma) \leq \dfrac{1}{k^2}$, equivalently

$$P(|X-\mu| > \epsilon) \leq \frac{\mathrm{Var}(X)}{\epsilon^2}$$

This is the one that proves the WLLN.
Jensen: g convex $\Rightarrow E[g(X)] \geq g[E(X)]$; g concave $\Rightarrow$ reverse. Square root is concave, hence $E(S) < \sigma$.

8. ORDER STATISTICS (needed whenever the MLE is $\max\{X_i\}$)

Let $X_{(n)} = \max\{X_i\}$ and $X_{(1)} = \min\{X_i\}$ from an i.i.d. sample.

$$F_{X_{(n)}}(x) = [F_X(x)]^n, \qquad f_{X_{(n)}}(x) = n\,[F_X(x)]^{n-1} f_X(x)$$

$$F_{X_{(1)}}(x) = 1 - [1 - F_X(x)]^n$$

$$f_{X_{(1)}}(x) = n\,[1 - F_X(x)]^{n-1} f_X(x)$$

Logic: the max is below x exactly when *all* n observations are, and independence turns that into a product. The min exceeds x exactly when all n do.
For Uniform$(0,\theta)$, $F_X(x) = x/\theta$ on $(0,\theta)$, so

$$f_{X_{(n)}}(x) = \frac{n x^{n-1}}{\theta^n}, \quad E\big[X_{(n)}\big] = \frac{n}{n+1}\theta$$

so $\max\{X_i\}$ is biased downward, and $\frac{n+1}{n}\max\{X_i\}$ corrects it.

9. READING A DENSITY (do this before anything else)

1. Find the support. The phrase "and zero otherwise" matters. Every integral runs over the support only.
2. Check whether the support involves θ. If it does, flag it now: the CRLB, the score identity and calculus-based MLE all become suspect (Sheet 1, box 13).
3. Identify the parameter and its role. Is θ the mean, the reciprocal of the mean, a variance, or a standard deviation? This decides whether the answer is $\overline{X}$, $1/\overline{X}$, $\widetilde{S}^2$ or $\widetilde{S}$.
4. Separate the normalizing constant from the part that depends on x; the constant often carries all of the θ dependence.
5. Verify $\int f = 1$ if you are unsure you copied it correctly.

10. SUMS AND CLOSURE PROPERTIES

For independent X and Y, use mgfs: $M_{X+Y}(t) = M_X(t)M_Y(t)$, then identify by uniqueness.
Closed under addition (independent case):
$N(\mu_1,\sigma_1^2) + N(\mu_2,\sigma_2^2) = N(\mu_1+\mu_2,\ \sigma_1^2+\sigma_2^2)$
Poisson(λ_1)+Poisson(λ_2) =Poisson$(\lambda_1+\lambda_2)$
$\chi^2_{(r_1)} + \chi^2_{(r_2)} = \chi^2_{(r_1+r_2)}$
Gamma(α_1,β)+Gamma(α_2,β) =Gamma$(\alpha_1+\alpha_2,\beta)$
Binomial(n_1,p)+Binomial(n_2,p) =Binomial(n_1+n_2,p)
Sum of n independent Exponential(mean θ) is Gamma(n,θ).
Not closed: Uniform, t, F.

CHEAT SHEET 2 • SIDE B

DISTRIBUTIONS, ALGEBRA, AND LEFTOVERS

lowest priority: he supplies pdfs, $E(X)$ and $\mathrm{Var}(X)$ on the exam

Per his email, any pdf, mean or variance you need will be given to you. This page is insurance and a recognition aid, not something to spend exam time on. If you run out of room while transcribing, cut here first.

8. DISTRIBUTION TABLE

Name	pdf / pmf	$E(X)$	$\mathrm{Var}(X)$	$I(\theta_0)$
Bernoulli(p)	$p^x(1-p)^{1-x}$	p	$p(1-p)$	$\frac{1}{p(1-p)}$
Binomial(n,p)	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$	—
Geometric(p)	$p(1-p)^{x-1}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	—
Poisson(λ)	$\frac{\lambda^x e^{-\lambda}}{x!}$	λ	λ	$\frac{1}{\lambda}$
Uniform(a,b)	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	—
Uniform$(0,\theta)$	$\frac{1}{\theta}$	$\frac{\theta}{2}$	$\frac{\theta^2}{12}$	*fails*
Expon., mean θ	$\frac{1}{\theta}e^{-x/\theta}$	θ	θ^2	$\frac{1}{\theta^2}$
Expon., rate λ	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{1}{\lambda^2}$
Gamma(α,β)	$\frac{x^{\alpha-1}e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha}$	$\alpha\beta$	$\alpha\beta^2$	—
$N(\mu,\sigma^2)$, σ^2 known	$\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$\frac{1}{\sigma^2}$
$N(0,\sigma^2)$, for σ^2	as above with $\mu=0$	0	σ^2	$\frac{1}{2\sigma^4}$
Beta(α,β)	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	—
$\chi^2_{(r)}$	Gamma$(r/2,2)$	r	$2r$	—
$t_{(p)}$	ratio, Sheet 1 box 8	0 $(p>1)$	$\frac{p}{p-2}$ $(p>2)$	—
$F_{(p,q)}$	ratio, Sheet 1 box 8	$\frac{q}{q-2}$ $(q>2)$	—	—
$\theta x^{\theta-1}$, $(0,1)$	as written	$\frac{\theta}{\theta+1}$	—	$\frac{1}{\theta^2}$

Relationships. $\chi^2_{(r)} = \text{Gamma}(r/2,2)$. Exponential $=$ Gamma$(1,\beta)$. A squared $N(0,1)$ is $\chi^2_{(1)}$. A sum of r independent $\chi^2_{(1)}$ is $\chi^2_{(r)}$. A sum of independent normals is normal.

9. LOG, EXPONENT AND DERIVATIVE RULES (for MLE work)

$\ln(ab) = \ln a + \ln b$; $\ln(a/b) = \ln a - \ln b$; $\ln(a^b) = b\ln a$
$\ln(e^u) = u$; $\ln 1 = 0$; $\ln(1/a) = -\ln a$

$$\frac{d}{d\theta}\ln\theta = \frac{1}{\theta}; \qquad \frac{d}{d\theta}\ln(1-\theta) = \frac{-1}{1-\theta}$$
$$\frac{d}{d\theta}\theta^{-1} = -\theta^{-2}; \qquad \frac{d}{d\theta}\theta^{-2} = -2\theta^{-3}$$
$$\frac{d}{d\theta}e^{-\theta} = -e^{-\theta}; \qquad \frac{d}{d\theta}\sqrt{\theta} = \frac{1}{2\sqrt{\theta}}$$
$$\frac{d}{d\theta}\theta^2 = 2\theta; \qquad \frac{d}{d\theta}\frac{1}{\theta^2} = -\frac{2}{\theta^3}$$

When the parameter is σ^2, set $v=\sigma^2$ and differentiate in v:

$$\frac{d}{dv}\ln v = \frac{1}{v}, \quad \frac{d}{dv}\frac{1}{2v} = -\frac{1}{2v^2}, \quad \frac{d}{dv}\frac{1}{2v^2} = -\frac{1}{v^3}$$

10. SUMMATION FACTS

$\sum_{i=1}^n X_i = n\overline{X}$; $\sum_{i=1}^n (X_i - \overline{X}) = 0$
$\sum_{i=1}^n c = nc$ for a constant c
$\sum(X_i-\overline{X})^2 = \sum X_i^2 - n\overline{X}^2$
Pull constants out of sums before differentiating; terms without θ vanish under $\partial/\partial\theta$.

11. SET THEORY AND BASIC PROBABILITY (compressed)

$P(A) \geq 0$; $P(S)=1$; disjoint $A_i \Rightarrow P(\cup A_i) = \sum P(A_i)$
$P(A\cup B) = P(A) + P(B) - P(A\cap B)$
$P(A^c) = 1 - P(A)$; $P(\emptyset) = 0$; $A \subset B \Rightarrow P(A) \leq P(B)$
De Morgan: $(A\cup B)^c = A^c \cap B^c$; $(A\cap B)^c = A^c \cup B^c$

$$P(A \mid B) = \frac{P(A\cap B)}{P(B)}$$

Bayes: $P(A \mid B) = \dfrac{P(B \mid A)P(A)}{P(B)}$, with $P(B) = \sum_i P(B \mid A_i)P(A_i)$ over a partition $\{A_i\}$
Independent events: $P(A\cap B) = P(A)P(B)$

12. CDF FACTS

$F_X(x) = P(X \leq x)$: non-decreasing, right-continuous, $F(-\infty)=0$, $F(\infty)=1$

$$f_X(x) = \frac{dF_X(x)}{dx} \text{ for continuous } X$$

$P(a < X \leq b) = F(b) - F(a)$
A pdf may exceed 1; a pmf may not. What must equal 1 is the total integral or sum.

13. EXAM PROCEDURE REMINDERS

- Say "by independence" wherever a variance of a sum splits.
- Say "identically distributed" wherever $E(X_i)$ becomes μ.
- Name the theorem: Thm 5.2.6, Thm 5.3.1(c), invariance property of the MLE, Slutsky, delta method, asymptotic properties of the MLE.
- Evaluate every identity at θ_0.
- Give the form requested; when unsure, write both the limiting and the asymptotic distribution.
- Label estimators $\widehat{\theta}_{MLE}$ or $\widehat{\theta}_{MOM}$.
- Answer a yes/no question with "Yes" or "No" first, then justify.
- A labeled setup on an unfinished problem beats a blank.

14. GAMMA FUNCTION AND USEFUL LIMITS

$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1}e^{-t}dt$
$\Gamma(\alpha+1) = \alpha\,\Gamma(\alpha)$; $\Gamma(n) = (n-1)!$ for integer $n \geq 1$
$\Gamma(1) = 1$; $\Gamma(1/2) = \sqrt{\pi}$

$$\lim_{n\to\infty}\frac{n-1}{n} = 1, \quad \lim_{n\to\infty}\frac{c}{n} = 0, \quad \lim_{n\to\infty}\frac{c}{n-1} = 0$$
$$\lim_{n\to\infty}\frac{n}{n+1} = 1, \quad \lim_{n\to\infty}\left(1+\tfrac{a}{n}\right)^n = e^a$$

Anything of the form $\frac{\text{constant in } n}{\text{polynomial in } n}$ goes to zero, which is what almost every consistency proof needs.

15. WHERE THINGS LIVE IN THE OTHER SHEET

Pipeline, score, $I(\theta_0)$, CRLB → Sheet 1 box 1
MLE steps and invariance → Sheet 1 box 2
MOM steps → Sheet 1 box 3
Delta method → Sheet 1 box 4
UMVUE test → Sheet 1 box 5
Consistency ritual → Sheet 1 box 6
Proof skeletons → Sheet 1 box 8
χ^2, t, F construction → Sheet 1 box 8 (side B)
Sampling distribution facts → Sheet 1 boxes 9, 9b
MSE → Sheet 1 box 10
WLLN, CLT, Slutsky, MLE theorem → Sheet 1 boxes 11, 11b
Score and information identities → Sheet 1 box 12
Regularity failure → Sheet 1 box 13